SRE / sysreseval Documentation

IUT d'Orsay, Université Paris-Saclay

UTILISATION

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CHAPTER

ONE

OVERVIEW

SysResEval (or **SRE**, for *Système Réseaux Évaluation*) is a software suite for managing exercise sessions and evaluations in network and system administration courses. It is built on top of the Kathara framework, using Docker containers and VDE networking.[1]

It was developed at **IUT d'Orsay, Université Paris-Saclay** (France).

SysResEval can be used by students autonomously, in supervised exercise sessions, or in time-limited evaluations.

Students interact only with a GUI, `sysreseval`, which calls the `sre` CLI on their behalf through the small `sre-wrapper` setuid helper The instructor can also use the CLI to set up an exam, start a project for a student, connect to a machine, launch an eval, etc…

1.1 Student Workflow

The student starts the **sysreseval** GUI, then chooses and opens a *lab*.[2] Each lab presents several tabs:

- **Schema** — the network topology. By default, machines that disallow connection are drawn in red, but every lab can override colors and shapes for machines and networks. Some machines may be hidden from the diagram entirely.
- **Informations** — general background and instructions, written in Markdown.
- **Questions** — a list of items the student must complete. Each question is one of:
  - a Markdown text describing a task to perform on the machines (e.g. *configure the network on m1*),
  - a form whose fields capture specific facts (e.g. *what is the MTU of `eth0` on m1?*); fields may be free-text validated by a regexp (e.g. an IPv4 address), a dropdown, or a checkbox,
  - a free-form multi-line text answer.
- **Terminals** — an embedded shell for each machine the student is allowed to connect to. A lab may expose these as plain root shells or, for example, as a `login` prompt that restricts students to a particular user account.
- **Machines** — a status table for each machine: state, NAT network, exposed ports. The students can use it to launch separate terminals sessions to a machine (useful when several terminal sessions are needed on one machine)
- **Evaluations** — lets the student trigger an automated evaluation of their work and view the resulting grade table.

[1] A first version was made with Marionnet and then rebuilt from scratch on Kathara / Docker.
[2] The word *lab* comes from Kathara (and Netkit). We use *lab* and *project* interchangeably in this documentation.

- **Apply Configuration** — lets the student put the project into a predefined state, for example a partial correction.

The **Questions**, **Evaluations**, and **Apply Configuration** tabs are optional and are shown only when the lab defines them.

Multiple labs can be open at once, letting a course be structured by topic instead of by session.

1.2 Monitoring of a session

During an exercices session, an instructor with the `sre watch` tool can monitor a whole class:

- consult the last evaluation result of any student (evaluations run periodically — the interval is a per-project parameter, typically one minute — even when the student does not trigger a self-evaluation).
- be alerted about any error during evaluations or by the lack of a recent evaluation
- display a list of all "grade elements" and the max, min and average grade of the students

The grade elements shown to the instructor can differ from those the student sees during a self-evaluation. For example, on a static routing exercise, the instructor may see detailed items (IPv4 address, default route, indirect route toward network A, etc.) while the student only sees a single "Configuration of machine m1".

1.3 Running exams

sysreseval manages time-limited exams. The instructor configures the exam and sysreseval starts each student's project — either immediately or at a scheduled time — displays a countdown, runs periodic evaluations (by default every 60 seconds), and shows a banner when the allotted time is up.

The instructor can change the duration during the exam, or even after it ends — the virtual machines are not stopped at the end (in practice, it is often easier to set a single duration for the whole class with a `dsh` call, then adjust it individually for students with special accommodations)

The student's final mark is the best overall grade across their evaluations.

1.4 Post-exam workflow

Each evaluation produces an archive file containing all of its data, readable via `sre cat`. If a student questions their score, the instructor can give a full explanation — for example, for a question about routing, by showing the output of `ip route` on m1 to reveal which routes were in place at the time. During exams, all terminal sessions are also recorded as asciinema files.

After the exam, the instructor can re-run the evaluation script — for instance to adjust the relative weight of items, or to fix a bug in the script — and then generate a per-student PDF with a detailed item-by-item breakdown, plus a recap ODS spreadsheet.

CHAPTER

TWO

EXAM ADMINISTRATION

2.1 Lifecycle at a glance

The model is **declarative**: the instructor writes a single file, `/var/lib/sre/exam.json`, and the GUI on each student PC does the rest. Two invariants make this work:

- **`exam.json` is the trigger.** Its presence puts the GUI in exam mode; its contents determine the current phase, the countdowns, and what to fire next. The instructor's only job is to create, edit, or delete this file — via `sre set-exam` and `sre del-exam`. The GUI polls it once per second and fires `pre-start-exam`, `start-exam`, `eval-exam`, and `end-exam` at the appropriate moments.
- **Every exam command is idempotent.** Each one can be re-run safely, which is what lets the GUI recover from transient failures and react to mid-exam edits. The instructor can extend the duration, add or remove a lab, or push back the end time at any point by editing `exam.json` (via `sre set-exam`); the change propagates within one second.

```
[Instructor]                          [GUI on each student PC, polling exam.json @ 1 Hz]
     |                                          |
     |  sre set-exam --labs tp1 \               |
     |    --start-after 09:00 \                 |  ← exam.json appears
     |    --end-before  11:00 \                 |
     |    --duration    120                     |
     |    --eval-interval 60                    |
     |                                          |
     |                              Phase: WAITING
     |                              • shows logo + countdown to 09:00
     |                              • ~60 s before start_after, sysreseval GUI fires
     |                                sre pre-start-exam (pulls images,
     |                                starts containers, writes pre_start_date)
     |                                          |
     +-- 09:00 ------------------------------->|
     |                              Phase: ACTIVE
     |                              • once projects are up, sysreseval GUI fires
     |                                sre start-exam (stamps started_at)
     |                              • shows remaining-time countdown
     |                              • fires sre eval-exam every 60 s
     |                              • student works, may self-grade
     |                                          |
     +-- 11:00 (or started_at + 120 min) ---->|
     |                              Phase: ENDED
```

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```
    |                              • sysreseval GUI fires sre end-exam once
    |                                (final concurrent eval + stamps ended_at)
    |                              • shows "That's All Folks"
    |                                          |
    |  sre del-exam                            |
    |  (removes exam.json, wipes projects)     |
    |  sre sheet / outline / re-eval           |  ← post-exam grading
```

2.2 Setting up an exam — `sre set-exam`

```
# Minimal - one lab, hard end time. No start time means "open immediately".
sre set-exam --labs s4/tp_ssh --end-before 11:00

# With some more options example
sre set-exam \
  --labs s4/tp_ssh s4/tp_dhcp:hard \
  --start-after  2026-06-15T09:00 \
  --end-before   2026-06-15T11:00 \
  --duration     90 \
  --eval-interval 30
```

In this case, each student gets 90 minutes, but every exam ends at 11:00 — even for students who start after 9:30.

Behaviour:

- **Incremental updates.** Only the options provided on the command line are written; everything else is preserved. `sre set-exam --duration 150` extends an in-progress exam without touching `labs` or `start_after`.
- **First-time requirements.** If `exam.json` does not yet exist, `--labs` is required, and so is at least one of `--end-before` / `--duration` (otherwise the exam would never end).
- **Date formats.** Both `--start-after` and `--end-before` accept full datetimes (`2026-06-01T09:00`, `2026-06-01 09:00`, `2026-06-01T09:00:00`) and time-only (`09:00`, `09:00:00`) — time-only is combined with today's date.
- **Lab validation.** Each lab name is checked against `sre list` (`get_lab_list(include_exam_only_labs=True)`); absolute paths must lie under `params.authorized_src_dir` (`/opt/sre/lab/` or `/home/admin1/`, etc..).
- **Resetting `start_after` resets state.** Changing `--start-after` clears `pre_start_date` and `started_at` so the GUI re-triggers `pre-start-exam` and `start-exam` for the new run.
- **Mis-exam edits.** Anything in `exam.json` may be changed while the exam is running; the GUI picks up changes within 1 s.
- **relative duration change**. During an exam, `--duration +X` or `--duration -X` add or substract X minutes.
-

Option	Effect
`--labs <lab[:fl ...>`	Authorised labs; `:flavor` selects a named `Flavor` preset (e.g. `tp1:hard`). Required when creating.
`--start- <dt>`	Exam opens after this moment. If omitted, the exam is considered open immediately.
`--end-be <dt>`	Exam closes after this moment, regardless of `started_at`.
`--durati <min>`	Hard cap on duration; the exam ends at `started_at + duration` *or* `end_before`, whichever fires first. Only effective once `started_at` is set. When updating an existing `exam.json` that already has a `duration` field, prefix the value with `+` or `-` to adjust the current duration (e.g. `--duration +30` adds 30 min, `--duration -15` subtracts 15 min).
`--eval-i <sec>`	Automatic-eval period (default 60).
`--record <bool>`	Record terminal sessions (`true`/`false`/`0`/`no`). Default `true`.

See *Exam Reference* for how the GUI drives the exam (phase computation and per-phase actions).

2.3 Monitoring an exam

Once each student machine mirrors its archives into a shared directory subtree on the instructor's machine (see the Installation chapter), `sre watch <directory>` provides a live dashboard of the exam:

- **Remaining time and latest grade** per student.
- **Inactivity alerts** when no new archive has appeared for a student in the last X seconds — usually a sign that `sysreseval` crashed or the machine was shut down.
- **Evaluation-error alerts** flagging any error raised by the last eval. A well-written lab should never produce errors; if a command is legitimately allowed to fail, pass `allow_error=True` to `Grade.cmd()` so the failure is not reported.

2.3.1 Clean up after an exam

```
sre del-exam
```

Removes `exam.json` and wipes every running project. Archive directories are preserved.

Note: in exam mode, quitting the GUI does not stop the Docker containers (unlike normal mode), so the instructor can keep them running afterwards — for instance to re-run a corrected version of the project if a bug surfaces during the exam.

2.4 Exam-only labs

Lab names containing any substring from `params.exam_only_affix = ["_EXAM_", "_OLD_", "_DRAFT_", "_TESTS_"]` are hidden from `sysreseval`'s lab picker. This is how you keep an exam lab invisible until it appears in `exam.json`:

- `/opt/sre/lab/s4/ssh_EXAM_final.py` — hidden from listings, but allowed in `--labs`.

- `/opt/sre/lab/_DRAFT_/work_in_progress.py` — hidden in any subdirectory whose name contains a flagged affix.

`sre set-exam --labs ...` and all the commands from privileged users (`sre start...`) accept hidden labs of course.

2.5 Post-exam workflow

2.5.1 Inspect a single archive

```
sre cat --grades --answers /var/lib/sre/archives/20260615110000_*.zst
```

Field option	Shows
`--grades`	Per-element grades, total grade, max
`--answers`	Student answers (incl. login, hostname, fullname, email, exam timing)
`--errors`	Errors raised during grading
`--tests`	Raw test results per machine (large)
`--data`	Serialized `Data` instance for the lab
`--files` / `--extract-files`	List or extract files saved into the archive
`--json`	One JSON dict per file on stdout (script-friendly)

2.5.2 Re-grade with a corrected project file

If a grading bug surfaces during the exam, fix-it and re-grade every archive:

```
sre re-eval -s /opt/sre/lab/_EXAM_/ssh_corrected.py \
            -p corrected \
            -d /tmp/regraded \
            -r /home/a/results/
```

Output archives appear in `/tmp/regraded/` with the `corrected` prefix. Use `sre check-eval -s ...` first to preview the diff without writing any file.

This only works if the corrected project file does not require any **new** command to be run on the virtual machines: re-evaluation replays the grading logic against the test output already stored in the archive, so a command that was never executed during the exam (say, `ip route` on `m1`) cannot be recovered after the fact. Within that limit, fixing a grading bug post-exam is straightforward.

To preview the impact of the correction on a single archive without writing anything, use:

```
sre check-eval --srelab corrected_project.py archive_file
```

which prints the diff between the stored grades and the re-graded ones.

The overall grade is computed by the `Grade0.mark_exo_eval()` method, which can be overridden to rescale or cap the final mark. For example, to multiply every mark by 1.1 and round up to the next 0.1, override `mark_exo_eval()` in the lab's `Grade` class:

```
def mark_exo_eval(self):
    if not self._grade_list or self._total_max == 0:
        return None
    import math
```

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```
    return min(
        self._maximum_mark,
        math.ceil(10 * 1.1 * self._maximum_mark * self._total_grade / self._total_max) /␣
↪10,
    )
```

2.5.3 Per-student PDF reports + summary spreadsheet

To create per-student PDF reports and a grade spreadsheet:

```
sre outline \
  -o /tmp/summary.ods \
  -d /tmp/reports \
  -r /var/lib/sre/archives/
```

Useful options:

- `--users-file <file>` — auxiliary `LOGIN NAME EMAIL` file (whitespace- or CSV-separated, `#` for comments) that supersedes the names and emails embedded in the archives. Handy when student logins are pseudonyms.
- `--remaining-time` — include each student's remaining exam time (from `exam_time_remaining` in the archive) in the summary.
- `--no-timeline` - omit the evaluation history table from PDF reports
- `--no-parts` - do not group PDF grade rows by GradePart (flat list, no subtotals)

2.5.4 Export to a spreadsheet

To get a full sheet of all grades elements of all evaluations with per-question summary:

```
sre sheet -o /tmp/exam-results.ods -r /var/lib/sre/archives/
```

CHAPTER

THREE

LAB AUTHORING GUIDE

A lab is a Python module that defines three classes: `Data`, `NetScheme`, and `Grade`, each inheriting from the corresponding `*0` base class in `lib_sre.py`. An optional `Flavor` class can be defined to parameterize the lab at start time.

A lab is usually a single `.py` file.

Directory labs

SRE also supports **directory labs** — a subdirectory containing a `srelab.py` plus per-state file trees (`initial/`, `state1/`, …) that are copied into the containers at the matching state transition. This is the Kathara layout, but in practice it's rarely needed: most setup is better expressed in code, since `Data`-derived values (random IPs, secrets, etc.) can then be injected into the generated files. The directory form is mainly useful when you need to ship payload that can't reasonably be generated at runtime — a pre-built apt package, a large data file, or anything else you don't want to download from the Internet on every lab start.

```
/opt/sre/lab/s4/tp_ssh/
+-- srelab.py          ← the module
+-- initial/           ← files injected at startup
|   +-- router/        ← files for the machine named "router"
|   |   +-- etc/network/interfaces
|   +-- all/           ← files copied to every machine
|       +-- etc/motd
+-- state1/            ← files injected when "sre state <lab> state1" is called
    +-- router/
        +-- etc/...
```

3.1 The authoring workflow on the CLI

While iterating on a `srelab.py`, you typically alternate between editing the module and exercising it with a handful of `sre` subcommands. Full reference lives in *CLI reference*; this section just names the commands you'll reach for as an author and explains the one option that exists for authors only.

3.1.1 `sre check [-p] <lab> [<state>]`

Static validation, no containers deployed. Imports the module, runs `Data.compute_pre_generate()` → `Data.generate()` → `data.compute_post_generate()`, instantiates `NetScheme`, applies `initial()` (op registration only), then calls `Grade.grade()` once. Catches import errors, missing fields, bad topology, and exceptions raised at registration time. With a trailing `<state>` argument, the named `@sre_state` method is also exercised. Run this first after every non-trivial edit — it's much faster than starting a real lab.

3.1.2 sre start --debug-project [-p] <lab> [--xauth-file <path>]

The author counterpart to `sre start`. Privileged only and incompatible with `--user`. Deploys the lab exactly like a normal start, then drops a `.private/debug_project` marker that the rest of the runtime keys on. Effect:

- **Topology**: every machine is shown in the GUI's *Machines* tab — even those declared `hidden=True` or `allow_connection=False`. *Connect* buttons are exposed for all of them so you can step inside infrastructure machines (DNS, monitors, helpers) to inspect their state.
- **Restrictions lifted**: user-mode guards on `sre state` and `sre connect` (which normally refuse to touch hidden machines or apply `user_allowed=False` states) are bypassed.
- **Grading**: when you `sre eval` a debug project, marks are always shown and every grade scope is surfaced — `no_mark_on_self_grade`, `hide_potential_penalty_grades_in_self_grade` and the periodic-eval gating are all ignored.
- **`-p` is implicit**: `--debug-project` always treats the lab argument as a filesystem path, so you can point it directly at the `.py` you're editing (e.g. `~/labs/draft.py`) without copying it into `/opt/sre/lab/` first.

The marker survives container restarts and is cleared when the project is `sre stop`ped or `sre wipe`d.

`--xauth-file <path>` is the other authoring-time option on `sre start`. By default, machines marked `x11_host=True` pick up the host's X magic cookie from the `SRE_XAUTH_COOKIE` environment variable (the sre-wrapper sets this for the GUI flow). When you start a project by hand from a shell where that variable isn't set — typical for `--debug-project` — pass `--xauth-file ~/.Xauthority` to source the cookie from a file instead. Privileged only.

3.1.3 sre connect <running_lab> <device> and sre exec <running_lab> <device> <command…>

`sre connect` opens an interactive shell inside `<device>` (using the machine's configured `shell`). `sre exec` runs a one-shot command and returns its stdout/stderr/exit code — useful for scripting checks against a running lab without holding a TTY. `sre exec` is privileged only; both honor the debug-project marker, so they can target hidden machines when the project was started with `--debug-project`.

`sre connect --exec <argument…>` is a hybrid: launches the machine's shell, runs the command, exits. Unlike `sre exec` it goes through the shell launcher and is available to students.

3.1.4 sre state <running_lab> <state_name>

Applies a non-`initial` `@sre_state` method against the live containers. The main use case during authoring is exercising the instructor's "fully-configured" state (often called `final`) so you can run `sre eval` against a known-good configuration and verify the grading code reports the expected full marks. In a non-debug project, students can only apply states declared with `@sre_state(user_allowed=True)` *and* only when the module sets `allow_user_states = True` — `--debug-project` lifts both guards.

3.1.5 sre eval <running_lab>

Runs an evaluation against the live containers — equivalent to pressing the *Start evaluation* button in the GUI. The evaluation also refreshes the project's view of the lab module, so any edits to `informations`, questions, or grade rubric take effect on the next display without needing to restart the lab.

3.1.6 `sre wipe`

Stops every running Kathara lab and removes everything under `/var/lib/sre/projects/`. The reset button between iterations: when a lab is misbehaving and `sre stop <running_lab>` isn't enough (orphaned bridges, broken Kathara state, leftover marker files), `sre wipe` brings the host back to a clean slate. It does **not** touch `exam.json` or archive directories.

A typical edit-test cycle:

```
sre check -p ~/labs/draft.py final            # validate module + the 'final' state
sre start --debug-project -p ~/labs/draft.py  # deploy, with debug visibility
sre connect <running_lab> router              # poke around
sre state <running_lab> final                 # apply the instructor's known-good state
sre eval <running_lab>                        # confirm grade == max
sre wipe                                       # reset before the next edit
```

3.2 Identifying a running lab and where its files live

Every running project lives in its own subdirectory of `/var/lib/sre/projects/`. The directory name encodes the launch timestamp, the lab name, and the student username:

```
/var/lib/sre/projects/{YYYYmmddHHMMSS}@@@{lab_name}@@@{username}/
```

e.g. `20260606143215@@@s4@ssh@@@alice`. List the directory to see what's currently running:

```
ls /var/lib/sre/projects/
```

Partial names are accepted. Every subcommand that takes a `<running_lab>` argument resolves it by substring match against the directory listing — you only need to type enough of the name to uniquely identify one project. If your substring matches several running labs, `sre` prints the candidates and exits so you can disambiguate; if it matches none, you get `no running lab matches '...'`. In practice the lab name itself (or even a fragment of it) is usually enough:

```
sre connect ssh router        # works if only one ssh project is running
sre connect 20260606 router      # disambiguate by timestamp prefix
sre connect alice router      # disambiguate by username
```

Inside the project directory you'll find:

Path (relative to the project dir)	Contents
`info.json`	Public metadata read by the GUI: lab title, `informations`, machine list, question list. Written at `sre start` and refreshed on `sre eval`.
`scheme.svg`	The topology diagram rendered with graphviz.
`answers/answers.json`	Student answers to `question_text` / `question_form` blocks.
`answers/cheat.json`	Optional instructor-provided cheat answers keyed by state name (used by `cheat_answers=`).
`shared/`	Bind-mounted into every container as `/home/sre/{running_lab_name}/` when the module sets `shared_path = True`. The host-side path for file exchange.
`.private/`	Mode `0o700`. Holds everything the runtime needs but students shouldn't touch.
`.private/data.json`	The serialized `Data` instance, reloaded on every later operation.
`.private/srelab`	Symlink to the lab's `.py` file (whatever `sre start` was pointed at).
`.private/files/`	Target of `self.cp_from_host()` / `self.cp_to_host()` — also where `cp_from_host` resolves relative source paths.
`.private/records/`	Asciinema recordings of terminal sessions, if `record_sessions` is enabled.
`.private/eval_in_progres`	PID lock created during `sre eval` to prevent concurrent evaluations.
`.private/debug_project`	Marker dropped by `sre start --debug-project` (see above).
`.private/auto_eval.log`	One line per student-triggered self-eval; drives the `self.auto_eval_count` counter exposed to `Grade.grade()`.

Evaluation archives are **not** stored under the project directory — they go to `/var/lib/sre/archives/` (plus any extra `archive_dirs` declared at module level), named `{YYYYmmddHHMMSS}_{running_lab_name}.zst`.

3.3 The four classes at a glance

A lab module defines up to four classes, each with a narrow role:

- `Data` — a `@dataclass` that holds every per-instance parameter of the lab (IP addresses, secrets, ports, randomized values). `Data.generate()` is called once at `sre start`; the resulting instance is serialized to `.private/data.json` and reloaded on every later operation.
- `Flavor` *(optional)* — a `@dataclass` that lets a single `srelab.py` produce different variants of the lab at start time (e.g. random vs. fixed addresses, easy vs. hard). The GUI renders `flavor_form` as a form; the chosen `Flavor` is then passed to `Data.generate(flavor)`.
- `NetScheme` — declares the network topology (`_machine_specs`, `_network_specs`, `_topology`) and the imperative configuration of each *state*. The `initial` state runs at `sre start`; additional `@sre_state` methods can be applied later with `sre state <lab> <name>`.
- `Grade` — registers what to evaluate and how to score it. It does **not** run anything itself: SRE drives a multi-step `run_tests()` loop that calls `Grade.grade()` repeatedly, executing the commands registered by each call between passes. See *The grading lifecycle*.

The flow at `sre start` is: `Data.generate(flavor)` → `NetScheme(data).initial()` (which lays out interfaces and writes files). The flow at `sre eval` is: reload `data.json` into a fresh `Data`, rebuild `NetScheme`, then run `Grade.run_tests()` against the live containers.

3.4 Data class

`Data` is a `dataclass` that holds all lab-specific parameters (IP addresses, random secrets, etc.). It inherits from `Data0`.

```
from dataclasses import dataclass
from ipaddress import IPv4Interface
from SRE.lib_sre import Data0
from ips import random_ipv4networks, random_ipv4s

@dataclass(slots=True)
class Data(Data0):
    secret: str = ''
    vlan_id: int = 0

    @classmethod
    def generate(cls, flavor=None):
        data = cls(secret="changeme", vlan_id=42)
        # data.nets and data.ips are injected automatically by Data0.__post_init__
        data.nets.lan, data.nets.mgmt = random_ipv4networks([24, 24],␣
↪from_private_network=True)
        ip = random_ipv4s(data.nets.lan, 1)[0]
        data.ips.router = IPv4Interface(f'{ip}/{data.nets.lan.prefixlen}')
        return data
```

Key points:

- `data.ips` holds `IPv4Interface` values (address + prefix length); `data.nets` holds `IPv4Network` values; `data.macs` holds `netaddr.EUI` (MAC address) values.
- All three containers are injected automatically into every `Data0` subclass instance by `__post_init__` — no declaration needed.
- The containers enforce their types: assigning a plain string to `data.ips` raises `TypeError`; always wrap with the correct type before assigning.
- `data.ips` requires `IPv4Interface` (not bare `IPv4Address`) — always assign with a prefix, e.g. `IPv4Interface('10.0.0.1/24')`.
- `Data.generate(flavor)` is called once at `sre start`; the result is saved to `data.json`.
- Serialization handles `IPv4Interface`, `IPv4Network`, `EUI`, and nested `Data0` subclasses transparently.

3.4.1 IP helper functions (from `/opt/sre/lib/ips.py`)

```
from ips import random_ipv4networks, random_ipv4s

# Returns a list of n disjoint networks with the given prefix lengths:
nets = random_ipv4networks([24, 28, 24], from_private_network=True)

# Returns n distinct random IPv4Address objects within a network:
hosts = random_ipv4s(nets[0], 3, exclude_nets=[nets[1]])
# Wrap as IPv4Interface before assigning to data.ips:
data.ips.host = IPv4Interface(f'{hosts[0]}/{nets[0].prefixlen}')
```

3.4.2 `compute_pre_generate` and `compute_post_generate`

Two optional lifecycle hooks let a `Data` subclass derive auxiliary values without storing them in `data.json`. Override either as needed; the defaults are no-ops.

Hook	Signature	Receive	Purpose
`compute_pr`	`@classmethod def compute_pre_generate(cls, flavor=None)`	class	Set **class-level** attributes derived from `flavor` (machine lists, topology sizes, role tables, etc.) before any `Data` instance exists.
`compute_po`	`def compute_post_generate(se`	instanc	Set **instance-level** attributes derived from the dataclass fields after generation or reload.

When they run:

1. At `sre start`: `Data.compute_pre_generate(flavor)` → `Data.generate(flavor)` → `data.compute_post_generate()`.
2. At `sre eval`, `sre state`, `sre connect`, etc.: after `data.json` is reloaded into a fresh `Data` instance via `from_json` / `from_dict` / `unpack`, both hooks run again — pre-generate first (with the persisted `flavor`), then post-generate.
3. `sre check` exercises the same sequence as a sanity check.

Because the hooks run on every reload, they should be **deterministic and side-effect-free**: no random number generation, no IP allocation, no file I/O. Anything that must be randomized once and then persisted belongs in `generate()`.

Typical use case — derive the machine list and per-machine spec from a `Flavor`:

```
@dataclass(slots=True)
class Data(Data0):
    secret: str = ''

    @classmethod
    def compute_pre_generate(cls, flavor=None):
        if flavor is None:
            flavor = Flavor()
        match flavor.network_size:
            case "small":
                r_max, m_max = 2, 2
            case "medium":
                r_max, m_max = 3, 4
            case _:
                r_max, m_max = 4, 7
        cls.routers = [f"r{i}" for i in range(1, r_max + 1)]
        cls.non_routers = [f"m{i}" for i in range(0, m_max + 1)]
        cls.machine_specs = {'gw': {'bridged': True}}

    @classmethod
    def generate(cls, flavor=None):
        data = cls(secret=random_password(16))
        # cls.routers / cls.non_routers are already populated by compute_pre_generate
        data.nets.lan = random_ipv4networks([24], from_private_network=True)[0]
        return data
```

`NetScheme.build()` and `Grade.grade()` can then read `self.data.routers` / `self.data.non_routers` directly, even on reload, without those fields needing to live in `data.json`.

3.5 `Flavor` class

A `Flavor` is an optional dataclass that parameterizes a lab at start time. If defined, the GUI presents a form to the student before starting.

A typical use case — see `lab/sre/static_routing.py` — is to switch a lab between two modes: when students work on it autonomously or during an exam, randomized IP addresses are preferable so each student has their own topology; in a supervised classroom, the instructor may want every student to share the same values, so a single explanation at the board applies to everyone. A `Flavor` lets the lab expose both modes from the same `srelab.py` and pick one at start time.

```
from dataclasses import dataclass
from typing import Tuple
from SRE.lib_sre import Flavor0

@dataclass(slots=True)
class Flavor(Flavor0):
    nb: int = 0

    flavor_form = """
    Number of clients: @@{nb:[0-9]+}@@
    Mode: @@{mode:>easy|hard}@@
    """

    def allowed_by_user(self) -> Tuple[bool, str]:
        """Return (True, '') if the student is allowed this flavor."""
        if self.nb <= 5:
            return True, ""
        return False, "Maximum 5 clients allowed."

# Named presets (accessible as Flavor.easy, Flavor.hard, etc.)
Flavor.easy = Flavor(nb=1)
Flavor.hard = Flavor(nb=5)
```

Module-level control:

```
flavor_form_at_startup = True   # show the flavor form when the student opens the lab
```

`Flavor` API:

Method	Description
`to_dict()` / `from_dict(d)`	Serialize/deserialize field values
`from_form_dict(d`	Build a `Flavor` from form field strings, coercing to declared types (`int`, `bool`, `float`, `str`). Extra keys are set as plain attributes.
`allowed_by_user(`	Returns `(bool, message)`. Override to restrict what students can choose.

In `Data.generate(flavor)`, check `if flavor is not None` before using flavor fields.

3.6 `NetScheme` class

`NetScheme` declares the lab's network topology and exposes the imperative API used by every state method. Topology is declared at class level via three dicts; per-state actions live in methods decorated with `@sre_state` (covered in the next section).

```
from SRE.lib_sre import NetScheme0, sre_state, make_tr

tr = make_tr('en')

class NetScheme(NetScheme0):
    # Declare machines: keys are machine names, values are Machine kwargs
    _machine_specs = {
        'router':  {'color': 'green'},
        'client':  {},
        'hidden':  {'hidden': True, 'allow_connection': False},
    }

    # Declare network display options
    _network_specs = {
        'lan':  {'color': 'yellow'},
        'mgmt': {'color': 'gray'},
    }

    # Declare topology: net_name → list of machines (or dict with explicit interface␣
↪numbers)
    _topology = {
        'lan':  ['router', 'client'],          # auto interface numbering
        'mgmt': {'router': 0, 'hidden': 1},    # explicit interface numbers
    }

    def __init__(self, data, running_lab_name):
        super().__init__(data=data, running_lab_name=running_lab_name)

        # Markdown text for the Informations tab (supports make_tr for i18n)
        self.informations = tr(
            "## Lab description\nConfigure routing.",
            fr="## Description\nConfigurez le routage.",
        )
```

3.6.1 Declarations and accessors

Attribute / Method	Description
`_machine_specs`	Class-level dict: `{name: {Machine kwargs}}`
`_network_specs`	Class-level dict: `{net_name: {display kwargs}}`
`_topology`	Class-level dict: `{net_name: [machine,...]}` or `{net_name: {machine: iface_index}}`
`self.data`	The `Data` instance
`self.informations`	Markdown text (or `TranslatedText`) for the Informations tab
`self.get_machines()`	Iterator over all `Machine` objects
`self.get_machine_names()`	Iterator over machine name strings
`self.get_networks()`	Iterator over all `Network` objects

3.6.2 `Machine` constructor parameters

Parameter	Default	Description
`image`	`params.default_doc`	Docker image
`bridged`	`False`	Enable host-network bridging (required for `ports`)
`x11_hos`	`False`	Forward X11 to the host: inject `SRE_HOST_IP` (and the host's X magic cookie as `SRE_XAUTH_COOKIE`) so GUI apps launched in the container display on the host's X server. Requires `bridged=True` (refused at startup otherwise)
`hidden`	`False`	Hidden from the Machines tab
`allow_c`	`True`	Show *Connect* button in the GUI
`shell`	`None`	Shell launched by `sre connect` (interactive terminal session)
`kathara`	`None`	Shell Kathara uses to execute startup `exec_commands` inside the container
`privile`	`None`	Run the container in privileged mode. Refused at startup unless `params.allow_privileged_machines = True`
`color`	`None`	Node color in the SVG diagram
`exec_co`	`[]`	Commands run at container start
`sysctls`	`{}`	Kernel parameters
`envs`	`{}`	Environment variables
`ports`	`[]`	Port mappings, e.g. `["80XX:80/tcp"]` (X = wildcard digit)
`ulimits`	`{}`	ulimit settings
`volumes`	`{}`	Volume mounts

Port wildcards: `"80XX:80/tcp"` allocates the first free port in 8000–8099.

3.7 State methods

A *state* is a method of `NetScheme` decorated with `@sre_state`. The body of the method does not execute commands directly — instead, it **registers** operations (run a shell command, write a file, copy a file from the host, …) which SRE then applies in order against the running containers.

The `initial` state is always applied at `sre start` and is where the lab configures interfaces, launches services, and writes per-machine files derived from `Data`. Additional states are applied on demand with `sre state <running_lab> <state_name>`. Their main use case is the instructor's workflow: a state that brings the project into a fully-configured, all-questions-answered shape, so the grading code can

be exercised end-to-end without going through the student workflow. Students see and can apply a state only if the module-level attribute `allow_user_states = True` **and** that state was decorated with `@sre_state(user_allowed=True)`.

```
@sre_state(user_allowed=False)
def initial(self):
    """Always applied at sre start."""
    d = self.data
    self.cmd('router', f'ip addr add {d.ips.router} dev eth0')
    self.cmd('router', 'sysctl -w net.ipv4.ip_forward=1')
    self.file('client', '/etc/resolv.conf', f'nameserver {d.ips.router.ip}\n')

@sre_state(user_allowed=True, description=tr("Final config", fr="Configuration finale"))
def final(self):
    """Optional additional state; students can apply it via 'sre state'."""
    self.cmd('router', 'iptables -A FORWARD -j ACCEPT')
```

3.7.1 `@sre_state` decorator

Parameter	Description
`user_allowed`	If `False`, students cannot apply this state themselves (only root/instructor can).
`description`	Human-readable label shown in the GUI (supports `TranslatedText` from `make_tr`).

3.7.2 Per-machine operations

These all register an op against a single machine and accept a `step=` parameter (see *Multi-step state setup* below).

Method	Description
`self.cmd(machine, command, step=1)`	Run a shell command inside `machine`.
`self.file(machine, path, content, permissions=0o644, owner="root:root", mtime=None, step=1)`	Create or overwrite a file inside `machine`. `content` may be `str` or `bytes`.
`self.append_to_file(machine, path, content, permissions=None, owner=None, mtime=None, step=1)`	Append to a file (creates it if missing).
`self.idempotent_append_to_file(machine, path, content, ..., step=1)`	Same as `append_to_file` but only appends if the file does not already end with `content` — safe to call repeatedly.

3.7.3 File transfer between host and container

Method	Description
`self.cp_from_host(src, machine, dest, owner="root:root", permissions=None, mtime=None, step=1)`	Copy a file from the host into `machine`. Relative `src` is resolved against `params.files_dir(running_lab_name)`.
`self.cp_to_host(machine, path, dest, step=1)`	Copy a file from `machine` back to the host, inside `params.files_dir(running_lab_name)`. `dest` is restricted to that directory.

3.7.4 Host-side operations

Method	Description
`self.host_cmd(command, step=1)`	Run a shell command on the **host** (not inside any container). Refused if `params.execute_commands_on_host is False`.
`self.host_callback(callable step=1)`	Invoke a Python callable on the host at this step, with no arguments. Useful when the next steps need values computed in Python.

3.7.5 Multi-step state setup — the `step` parameter

Every state-method operation accepts `step=N` (default `1`). When SRE applies a state it groups all registered ops by step and runs them in ascending order: every step-1 op finishes before any step-2 op starts. This matters when later ops depend on earlier ones being in place — for example writing a config file at step 1 and only restarting the service at step 2:

```
@sre_state(user_allowed=False)
def initial(self):
    self.file('dns', '/etc/unbound/unbound.conf', conf, step=1)
    self.cmd('dns',  'systemctl restart unbound',    step=2)
```

Inside a single step the per-machine op order is preserved, and ops on different machines run in parallel. `host_cmd` / `host_callback` for step `N` run after all container ops of step `N` finish.

3.7.6 Network config helpers (from `/opt/sre/lib/net_config.py`)

```
from net_config import set_net_config_entry, set_sysctl, NetConfigEntry, SysctlConfig

# In initial():
set_net_config_entry(net_scheme=self, machine_name='router', nc_entry=[
    ([data.ips.router_lan], [(IPv4Network('0.0.0.0/0'), data.ips.gw)]),
])
set_sysctl(net_scheme=self, machine_name='router', sysctl_config={'ipv4.ip_forward': 1})
```

`NetConfig` is a list of interface entries; each entry is `([addresses], [(dest_network, gateway), ...])` or `'dhcp'` or `None`.

Persistent equivalents — these write to `/etc/network/interfaces` and `/etc/sysctl.d/99-sre.conf` so the config survives a container restart:

- `set_persistent_net_config_entry(net_scheme, machine_name, nc_entry)`
- `set_persistent_sysctl(net_scheme, machine_name, sysctl_config)`

3.7.7 State helpers (from `/opt/sre/lib/state_helpers.py`)

Convenience functions called inside `@sre_state` methods:

```
from state_helpers import (set_unbound_server, set_basic_unbound_server,
                           change_password, create_user,
                           hosts_file_content, create_hosts_file)
```

Function	Description
`set_unbound_server(net_s machine)`	Drop a permissive Unbound DNS config into `/etc/unbound/unbound.conf` and start the service. (Thin alias for `set_basic_unbound_server`.)
`set_basic_unbound_server machine)`	Same as above; explicit name when you want to make the "basic / permissive" intent obvious.
`change_password(net_sche machine, username, password)`	Set a user's password via `chpasswd`. The password is written to a temporary file (never passed on the command line).
`create_user(net_scheme, machine, username, password, uid=None, gid=None)`	Create a user with `useradd` (if not already present) and set its password. `uid`/`gid` are passed as integers. The password is written to a temporary file; the username is passed via an environment variable to prevent shell injection.
`hosts_file_content(net_s domain_extension, included=None, ips=None, separator="\ t\t")`	Return `/etc/hosts` lines for a set of machines. Each machine gets one line per network it is connected to (two lines if multi-homed, with `machine_net` naming). Addresses are read from `net_scheme.data.ips.*` by convention (`machine` for single-homed, `machine_net` for multi-homed), or from the `ips` dict if provided. `included` defaults to all visible machines.
`create_hosts_file(net_sc domain_extension, machine_list=None, included=None, ips=None, separator="\ t\t")`	Write `/etc/hosts` to each machine in `machine_list` (defaults to all visible machines). Each file begins with standard loopback entries (`127.0.0.1 localhost`, `127.0.1.1 <machine>`) followed by the lines from `hosts_file_content()`.

3.8 `Grade` class

`Grade` declares **what** the evaluator checks and **how** results are scored. It does not execute anything itself: every `self.test(...)` call merely registers a command. SRE then drives a multi-step loop (`run_tests()`) that calls `Grade.grade()` repeatedly, executing registered commands between passes and feeding their real results back into the next call.

```
from SRE.lib_sre import Grade0

class Grade(Grade0):
    def grade(self):
        """Register tests, questions, and grade elements.
        Called multiple times per evaluation - must be side-effect-free."""
        super().grade()

        # Register a test: run a shell command in a container
        result, code = self.test('router', 'ping -c1 -W1 8.8.8.8', timeout=5)

        # Register a question (student's text answer)
        answer = self.question_text(
            title='Default gateway',
            description='What is the default gateway for the client?',
            default_answer='',
        )

        # Register a grade element and set its score
        self.add_grade_element('Connectivity', max_grade=4)
```

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```
        if code == 0:
            self.set_grade('Connectivity', 4)
        elif '64 bytes' in result:
            self.set_grade('Connectivity', 2)

        self.add_grade_element('Gateway answer', max_grade=2)
        if str(self.data.ips.router.ip) in answer:
            self.set_grade('Gateway answer', 2)
```

3.8.1 The grading lifecycle

`Grade.grade()` is **not** called once. It is called repeatedly by `run_tests()`, with the registered commands actually executed between calls. The loop is:

1. `self.step` starts at 0, `self.max_step` at 1. `load_answers()` reads the student's `answers.json`.
2. `reset_before_grade()` clears questions, grade rubrics, and section counters.
3. `grade()` is called.
   - `self.test(machine, cmd, step=N)` registers `cmd` under `(machine, N)` and returns `(default_value, default_code)` (i.e. placeholders) the *first* time it is seen.
   - If a call uses `step=K` larger than `self.max_step`, `max_step` is bumped to `K`, extending the loop.
   - `self.add_grade_element` / `self.set_grade` / `self.question_*` calls all register entries; they don't read any container state directly.
4. `self.step` is incremented to `N`. All commands registered at step `N` are bundled per machine into one `EXETESTS@@@cmd1@@@cmd2@@@…` env var and run inside each container in parallel (16-worker `ThreadPoolExecutor`, via `/usr/local/sbin/exetests.py`). Host-side `test_host()` commands for step `N` run in parallel on the host.
5. Results are stored back into `self._tests[(machine, N)]`. The loop returns to step 2. On this pass, the registration calls for step `N` find the entry already populated and return the **real** `(stdout, exit_code)`; the code paths gated on those results now execute.
6. When `self.step > self.max_step`, the loop exits. The archive (zstd-compressed msgpack with all tests, answers, errors, and grade list) is written to `params.archives_dir` and to every directory in the module-level `archive_dirs`.

This has two consequences for `grade()`:

- **It must be side-effect-free.** It is run several times — at least twice (one registration pass, one result pass), more if you use multi-step tests. Never call `subprocess`, write to disk, or mutate global state inside it. All work goes through `self.test` / `self.question_*`.
- **`self.test()` returns placeholders the first time it is reached.** Code that branches on the return value (e.g. `if code == 0:`) executes both with the placeholder (no-op) and with the real result. Make sure the placeholder branch is harmless — it will execute, but its `add_grade_element` calls are wiped by `reset_before_grade()` before the next pass, so only the final pass's calls end up in the archive.

A worked two-step example: configure step 1 to perform a setup action, then read its effect at step 2.

```
def grade(self):
    super().grade()
    self.test('router', 'systemctl restart bird', step=1)
    out, code = self.test('router', 'birdc show route', step=2)
```

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```
    self.add_grade_element('OSPF routes', max_grade=5)
    if 'OSPF' in out:
        self.set_grade('OSPF routes', 5)
```

On the first call, both `self.test()` invocations register; both return placeholders. SRE executes the step-1 command (`systemctl restart bird`). On the second call, the step-1 test returns its real result; the step-2 test registers and returns its placeholder. SRE executes the step-2 command. On the third call, both tests return real results; `out` contains the routing table, the rubric is populated, and the loop exits.

3.8.2 Grade parts

A *grade part* is a named group that bundles related rubric items together. Parts are **purely presentational**: they affect how grades are displayed in the GUI's *Evaluations* tab and in the PDFs produced by `sre outline` — each part is rendered as a labeled block with a **subtotal row** summing its elements. The overall total, the archive contents, and the marks returned by `mark_exo_eval()` / `mark_self_eval()` are unchanged.

Register a part with `self.add_grade_part(title, description='')` and pass the returned object as `grade_part=` to every element that belongs to it:

```
def grade(self):
    super().grade()

    part1 = self.add_grade_part("part1", tr("Client DNS dig"))
    self.add_grade_element(title="dig_host1_a", max_grade=1, grade=int(host1_ok),
                           grade_part=part1, description=tr("dig - A de host1"))
    self.add_grade_element(title="dig_host2_a", max_grade=1, grade=int(host2_ok),
                           grade_part=part1, description=tr("dig - A de host2"))

    part2 = self.add_grade_part("part2", tr("Serveur DNS cache Unbound"))
    self.add_grade_element(title="unbound_running", max_grade=2, grade=int(unbound_ok),
                           grade_part=part2, description=tr("unbound actif"))
    # ...
```

Elements registered without `grade_part=` are shown ungrouped (above or between the parts, in registration order). Parts are rendered in the order they were registered with `add_grade_part()`. See `lab/sre/dns1.py` for a real-world example with three parts.

3.8.3 Self-eval vs instructor-eval scope

Every `add_grade_element` call takes a `scope=` bitmask that decides which audiences see the element. Three values are defined in `params`:

Value	Visible in
`SELF_EVAL_SCOPE` (1)	Student self-eval only — `sre eval --auto-eval` (GUI **Start evaluation** button)
`EXO_EVAL_SCOPE` (2)	Instructor view only — `sre eval` (non-auto), `sre outline`, `sre sheet`, periodic exam-mode evals
`BOTH_EVAL_SCOPE` (3, default)	Both audiences

The total mark reflects only the elements in the active scope: `mark_self_eval()` sums elements with `scope & SELF_EVAL_SCOPE`, `mark_exo_eval()` sums elements with `scope & EXO_EVAL_SCOPE`. The same `grade()`

body can therefore produce a different total for a student self-eval than for the instructor's archive.

A common pattern is to register a **detailed rubric** for the instructor and a **coarse summary** for the student — enough feedback to know something is broken in a given part, without disclosing which individual check failed:

```
from SRE import params

def grade(self):
    super().grade()

    part1 = self.add_grade_part("part1", tr("DNS resolution"))
    host1_ok = ...   # bool computed from self.test(...) results
    host2_ok = ...
    ptr_ok   = ...

    # Detailed rubric - only the instructor (and sre outline / sre sheet) sees these
    self.add_grade_element('dig host1 A',   max_grade=1, grade=int(host1_ok),
                           grade_part=part1, scope=params.EXO_EVAL_SCOPE)
    self.add_grade_element('dig host2 A',   max_grade=1, grade=int(host2_ok),
                           grade_part=part1, scope=params.EXO_EVAL_SCOPE)
    self.add_grade_element('dig host1 PTR', max_grade=1, grade=int(ptr_ok),
                           grade_part=part1, scope=params.EXO_EVAL_SCOPE)

    # Summary - what the student sees during self-eval
    all_ok = host1_ok and host2_ok and ptr_ok
    self.add_grade_element('part1_global', max_grade=1, grade=int(all_ok),
                           scope=params.SELF_EVAL_SCOPE,
                           description='Part 1 (combined)')
```

During the student's self-eval, only the `SELF_EVAL_SCOPE` element appears: one row per part with a pass/fail score (no need for grade parts there). During the instructor's evaluation, only the three `EXO_EVAL_SCOPE` rows appear. `scope` and `grade_part` compose freely — elements in the same part can have different scopes and will be subtotalled accordingly for each audience.

3.8.4 Cheat answers

`cheat_answers` on `question_text` / `question_form` maps a state name (e.g. `'final'`) to an answer value. When the student has applied that state, the cheat answer is used instead of the student's actual input. Useful when the instructor's `final` state should produce a fully-passing evaluation.

3.8.5 Letter grades vs numeric marks

By default, marks are numeric, scaled to `params.default_maximum_mark`, and rounded to one decimal. Set `self._use_numerical_marks = False` (per instance, inside `grade()` or earlier) to switch to letter grades on the **total**: `A+` 18/20, `A` 16, `B` 14, `C` 12, `D` 10, else `F`.

Per-element letter conversion (`OK` / `MEH` / `FAIL`) is also available via `GradeElement.to_grade_letter()` — full marks → OK, partial → MEH, zero → FAIL.

3.8.6 Overriding `mark_exo_eval()` to adjust the overall mark

The overall mark stored in the archive (and surfaced by `sre outline`, `sre sheet`, and `sre cat`) is produced by `Grade.mark_exo_eval()`. The default implementation is:

```
def mark_exo_eval(self):
    return self._compute_mark(self._total_grade_exo_eval, self._total_max_exo_eval)
```

i.e. `ceil(10 · maximum_mark · total_grade / total_max) / 10` in numerical mode, or the A+/A/B/C/D/F letter in letter mode. `_total_grade_exo_eval` / `_total_max_exo_eval` are the sums over every `add_grade_element` that contributes to the instructor's view (i.e. excluding self-eval-only rubrics). `mark_self_eval()` is the counterpart used during student self-evaluations.

Override `mark_exo_eval()` (and/or `mark_self_eval()`) on the `Grade` subclass when you want a non-default scale — typically because the maximum of the rubric does not match the intended denominator. Common reasons:

- **Fixed denominator.** The exam was designed against an absolute target (e.g. 39 points). The rubric may sum to more, but you want the mark expressed against that fixed value. Replace `_total_max_exo_eval` with the constant, and cap at `_maximum_mark` so bonuses don't push above the cap:

  ```
  import math

  class Grade(Grade0):
      def mark_exo_eval(self):
          if not self._grade_list or self._total_max_exo_eval == 0:
              return None
          return min(self._maximum_mark,
                     math.ceil(10 * self._maximum_mark * self._total_grade_exo_eval /␣
  ↪39) / 10)
  ```

- **Bonus / penalty handling.** When penalty rubrics are registered with `max_grade=0` (so they only subtract), they don't enlarge the denominator — but if you want them to *not* subtract below zero either, clamp `total_grade` before scaling.

`mark_exo_eval()` is called once at the end of `run_tests()`, after every pass of `grade()` and after `compute_total()`. By that time `self._grade_list`, `self._total_grade_exo_eval` / `_total_max_exo_eval`, `self._maximum_mark`, and `self._use_numerical_marks` are all final, so the override is free to read them. Return `None` to signal "no mark" (shown as blank), a `float` for a numeric mark, or a string for a letter grade.

3.8.7 Grade methods

Method	Description
`self.test(machine, command, step=1, timeout=20, allow_error=False, default_value='', default_code=0)`	Register a command in a container; returns `(stdout, exit_code)` (placeholder on first pass, real result on subsequent passes).
`self.test_host(command, step=1, timeout=20, allow_error=False, default_value='', default_code=0)`	Same as `self.test` but the command runs on the host. Refused if `params.execute_commands_on_host is False`.
`self.question_text(title, section='', description='', hash=None, order=None, default_answer='', cheat_answers=None)`	Register a free-text question; returns the student's answer (or `default_answer`).
`self.question_form(title, section='', description='', hash=None, order=None, cheat_answers=None)`	Register a form question with inline `@@{field:regex}@@` (text), '@@{field:>opt1
`self.question_dummy(title, section='', description='', hash=None, order=None)`	Display-only block (no input shown to the student).
`self.add_grade_part(title, description='')`	Register a named group of grade elements and return the resulting `GradePart`. Pass it to `add_grade_element(..., grade_part=...)` to associate elements with this group; parts render in registration order with a subtotal row per part in the GUI Evaluations view and in `sre outline` PDFs.
`self.add_grade_element(title, max_grade, description='', grade=0, scope=params.BOTH_EVAL_SCOPE, grade_part=None)`	Add a graded rubric item. `scope` is a bitmask: `SELF_EVAL_SCOPE` (1) restricts the element to student self-eval, `EXO_EVAL_SCOPE` (2) restricts it to instructor/auto eval / `sre outline` / `sre sheet`, `BOTH_EVAL_SCOPE` (3, default) shows it everywhere. `grade_part` is a `GradePart` returned by `add_grade_part()`.
`self.set_grade(title, grade)`	Set the score of a previously registered element.
`self.section(level=0, fmt=None, show=None, pad=None)`	Increment the section counter at `level` and return its formatted label (e.g. `"I."`, `"I.1."`) — for grouping questions under numbered headings.
`self.current_section(level=0, fmt=None, show=None, pad=None)`	Same formatting as `section()` but without incrementing — read the current label for the same level.
`self.add_error(error, category=ErrorCategory.ERROR, step=1)` / `self.add_warning(warning, step=1)`	Record an error/warning in the archive. Only registered when `self.step == step` so the same call doesn't fire on every pass. `add_warning` is a thin alias for `add_error(..., category=ErrorCategory.WARNING)`.
`self.data`, `self.step`, `self.max_step`, `self.net_scheme`	Instance attributes available inside `grade()`.
`self.auto_eval_count`	Number of student-triggered self-evaluations performed on this project *before* the current run (read from `.private/auto_eval.log`; `0` for the first self-evaluation, instructor evaluations, periodic background evals,

3.9 Grading Library Reference

3.9.1 DHCP helpers (from /opt/sre/lib/dhcp.py)

```
from dhcp import DhcpParameters, DhcpSubnet, set_dhcp_server, get_dhcp_server,␣
↪check_running_dhcp_server
```

Data classes:

Class	Purpose
DhcpSubne	One subnet block: subnet, range_start, range_end, optional routers, dns_servers, domain_name, default_lease_time, max_lease_time, fixed_addresses
DhcpParam	Full server config: interfaces_v4, interfaces_v6, subnets, authoritative, default_lease_time, max_lease_time, ddns_update_style

Functions used in NetScheme (state setup):

set_dhcp_server(net_scheme, machine, dhcp_params, step=1) — writes /etc/default/isc-dhcp-server and /etc/dhcp/dhcpd.conf from a DhcpParameters instance, then enables and restarts isc-dhcp-server.

Functions used in Grade (evaluation):

get_dhcp_server(grade, machine, step=1) → (DhcpParameters | None, int) — reads and parses the DHCP server configuration from the running container. Returns the parsed parameters and the number of parse errors. Returns (None, 1) if /etc/default/isc-dhcp-server is absent.

check_running_dhcp_server(grade, machine) → (bool, list[str]) — checks whether isc-dhcp-server is currently active. Returns (running, interfaces) where interfaces is the list of interface names dhcpd is bound to (e.g. ["eth0"]), or ["*"] if it listens on all interfaces. Interfaces are read from the live process command line, not from the config file.

3.9.2 TLS helpers (from /opt/sre/lib/tls.py)

```
from tls import eval_rsa_private_key, set_rsa_private_key, eval_self_signed_certificate,␣
↪eval_certificate
```

Functions used in NetScheme (state setup):

set_rsa_private_key(net_scheme, machine_name, key_file, password, bits=4096, cipher='AES-256-CBC') — generates an encrypted RSA private key inside the container using openssl genrsa. File paths and the password are shell-quoted automatically.

Functions used in Grade (evaluation):

eval_rsa_private_key(grade, machine_name, key_file, password=None, bits=4096, cipher='AES-256-CBC', step=1) → bool — verifies that key_file is an RSA private key with the expected size and PEM cipher. Returns True if all checks pass.

eval_self_signed_certificate(grade, machine_name, key_file, cert_file, password, cn=None, bits=4096, cipher='AES-256-CBC', step=1) → dict — checks that cert_file is a valid self-signed certificate whose public key matches key_file. Returns a dict with keys subject, issuer, common_name, not_before, not_after, serial, fingerprint.

eval_certificate(grade, machine_name, key_file, cert_file, ca_file, step=1) → dict — checks that cert_file is a valid certificate signed by ca_file and whose public key matches key_file. Returns the same dict as eval_self_signed_certificate.

3.9.3 TCP port helpers (from /opt/sre/lib/grade_helpers.py)

```
from grade_helpers import eval_tcp_server
```

`eval_tcp_server(grade, machine_name, port, step=1) → bool` — checks whether a process is listening on `port` (TCP) inside the container. Returns `True` if the port is bound.

3.9.4 OSPF helpers (from /opt/sre/lib/frr.py)

```
from frr import get_ospf_interfaces
```

`get_ospf_interfaces(grade, machine_name, step=1) → dict` — runs `vtysh -c "show ip ospf interface"` inside the container and parses the output. Returns a dict keyed by interface name, each value containing: `area`, `cost`, `state` (DR/BDR/DROther/…), `dr`, `bdr`, `neighbor_count`, `adj_neighbor_count`.

3.9.5 Standard machine wiring (from /opt/sre/lib/std.py)

`machine_config(net_scheme, machine_name, config)` — concise helper for wiring a machine inside `NetScheme.build()` (or `initial()`). `config` is a `(interfaces, sysctls)` tuple:

- `interfaces`: list of `(network_name, address, routes)` entries, where `routes` is a list of `(dest_network, gateway)` pairs.
- `sysctls`: dict of kernel parameter names to values.

Compared to calling `set_net_config` and `set_sysctl` directly, `machine_config` resolves network names through the `NetScheme` topology automatically.

3.9.6 Miscellaneous generators (from /opt/sre/lib/utils.py)

```
from utils import random_password, random_sentence
```

Function	Description
`random_password(len`	Returns a random alphanumeric password of the given length
`random_sentence(len`	Returns a random sequence of lowercase words, space-separated, approximately `length` characters long

3.9.7 Network inspection helpers (from /opt/sre/lib/net_config.py)

These functions are used inside `Grade.grade()` to read live container state.

```
from net_config import (get_ip_addresses, get_routes, get_sysctl_conf,
                        get_net_config_entry, get_persistent_net_config_entry,
                        eval_net_config, get_ip_forward, get_sys_parameter_bool,
                        get_sys_parameter)
```

Function	Returns	Description
`get_ip_addresses(grade, machine, step=1)`	`dict[str, list[tuple[str, int]]]`	Run `ip a`; return `{iface: [(addr, prefixlen), ...]}` (sorted by prefix desc, addr asc)
`get_routes(grade, machine, step=1)`	`dict[tuple[str, int], tuple[str, str,int]]`	Run `ip route`; return `{(net, mask): (via, dev, metric)}` — `default` maps to `('0.0.0.0', 0)`
`get_sysctl_conf(grade, machine, step=1)`	`dict[str, str]`	Read `/etc/sysctl.conf` and `/etc/sysctl.d/*.conf`; return `{key: value}`
`get_ip_forward(grade, machine, step=1)`	`bool`	Read `/proc/sys/net/ipv4/ip_forward`; return `True` if `1`
`get_sys_parameter(grade, machine, param, step=1)`	`str | None`	Read an arbitrary `/proc/sys/...` file (e.g. `'net.ipv4.ip_forward'`); return string value or `None`
`get_sys_parameter_bool(grad machine, param, step=1)`	`bool | None`	Same as `get_sys_parameter` but cast to `bool` (`'1'` → `True`, `'0'` → `False`, else `None`)
`get_net_config_entry(grade machine, step=1)`	`NetConfigEntry`	Run `ip a` + `ip route`; reconstruct a `NetConfigEntry` from live state
`get_persistent_net_config_e machine, step=1)`	`tuple[NetConfigEntry int]`	Parse `/etc/network/interfaces`; return `(entry, n_errors)`

`eval_net_config(grade, expected, machine_name=None, current=None, step=1)` — compare a live or provided `NetConfigEntry` against an expected one. Returns an attribute-accessible dict with the following keys:

Key	Description
`ips`	Number of IP addresses that match
`ips_expected`	Number of IP addresses in expected
`default_route`	`1` if the default route matches, `0` otherwise
`default_route_expected`	`1` if expected has a default route
`other_routes`	Number of matching non-default static routes
`other_routes_expected`	Number of non-default routes in expected
`wrong_routes`	Non-default routes present in current but not in expected
`dhcp_interfaces`	Number of positions where both expected and current are `'dhcp'`
`dhcp_interfaces_expected`	Count of `'dhcp'` entries in expected
`none_interfaces_expected`	Count of `None` entries in expected

If `current` is `None`, `get_net_config_entry(grade, machine_name, step)` is called automatically.

3.9.8 Ping helper (from `/opt/sre/lib/ping.py`)

```
from ping import eval_ping
```

`eval_ping(grade, src, dest, step=1, net_config=None) → bool` — run `ping -c 1 -w 1 dest` from `src` machine; return `True` if `"bytes from"` appears in the output.

`src` and `dest` can each be:

- An `IPv4Address` or valid IPv4 string — used directly; `src` is resolved by reverse-lookup in `net_config`
- `"machine_name"` — machine name looked up in `net_config`; `dest` → first interface IP
- `"machine_name:N"` or `"machine_name:ethN"` — resolves to interface index N of that machine

If `net_config` is not provided, `grade.net_scheme.net_config` is used. Raises `ValueError` on resolution failure.

3.9.9 SSH helpers (from `/opt/sre/lib/ssh.py`)

Used in `@sre_state` methods (state setup):

```
from ssh import (add_ssh_monitor_agent, create_ssh_key_on_host,
                 remove_ssh_password_authentication_on_sshd,
                 set_forward_ssh_agent_in_ssh_config, copy_ssh_pub_key_on_machine)
```

Function	Description
`add_ssh_monitor_agent(net_sc machine, step=1)`	Deploy and start an SSH monitor daemon on `machine`. It tails `/var/log/auth.log` and writes one line per forwarded-agent key to `/var/log/.ssh_monitor.log` when a user logs in via SSH. Used with `eval_ssh_connection_with_ssh_agent()`.
`create_ssh_key_on_host(net_s filename, bits=4096, key_type='rsa', password=None, step=1)`	Generate an SSH key pair on the **host** (`ssh-keygen`). Returns the `filename`.
`remove_ssh_password_authenti machine, restart_ssh=False, step=1)`	Set `PasswordAuthentication no` in `/etc/ssh/sshd_config`. Optionally restart sshd.
`set_forward_ssh_agent_in_ssh machine_name, step=1)`	Write `/etc/ssh/ssh_config.d/forward_agent.conf` with `ForwardAgent yes` for all hosts.
`copy_ssh_pub_key_on_machine( machine, pub_key, username, step=1)`	Copy a public key file (host path) into `~username/.ssh/authorized_keys` on `machine`.

Used in `Grade.grade()` (evaluation):

```
from ssh import (check_ssh_key, eval_ssh_public_key_in_authorized_keys,
                 eval_ssh_agent_exists, eval_ssh_agent_with_loaded_key,
                 eval_ssh_possible_with_password_authentification,
                 eval_ssh_connection_with_password, eval_ssh_connection_with_key,
                 eval_ssh_connection_with_ssh_agent, eval_synchronized_file)
```

Function	Returns	Description
`check_ssh_key(grade, machine, private_key, key_type='rsa', bits=4096, password=None, step=1)`	`bool`	Verify that a key pair on `machine` has the expected type, bit length, and passphrase. Public key expected at `private_key + '.pub'`.
`eval_ssh_public_key_in_authorize`	`bool`	Check that a public key appears in `~username/.ssh/authorized_keys` with correct ownership and restricted permissions. Provide either `public_key_file` (host path) or `public_key` (content string).
`eval_ssh_agent_exists(grade, machine_name, username, step=1)`	`bool`	Return `True` if an `ssh-agent` process is running as `username` on `machine_name`.
`eval_ssh_agent_with_loaded_key(g`	`bool`	Check that a specific private key (on the host) is loaded in a running agent belonging to `username` on `machine_name`.
`eval_ssh_possible_with_password_`	`bool`	Probe `dest_machine`'s sshd from `src_machine`; return `True` if password authentication is offered.
`eval_ssh_connection_with_passwor`	`bool`	Read `/var/log/auth.log` on `machine_name`; return `True` if a successful password login for `username` is present.
`eval_ssh_connection_with_key(gra`	`bool`	Read `/var/log/auth.log`; return `True` if a successful public-key login for `username` is present.
`eval_ssh_connection_with_ssh_age`	`bool`	Read `/var/log/.ssh_monitor.log` (written by `add_ssh_monitor_agent`); return `True` if `username` logged in using an agent carrying `key_on_host`.
`eval_synchronized_file(grade, machine_list, filename, step=1)`	`tuple[` `str]`	Return `(True, content)` if `filename` has identical content and mtime on every machine in `machine_list`; `(False, '')` otherwise.

3.9.10 Packet capture helpers (from `/opt/sre/lib/pcap_gen.py`)

Used in `@sre_state` methods (state setup):

```
from pcap_gen import generate_pcap_tcp_example, setup_tcp_client_server
```

`generate_pcap_tcp_example(net_scheme, src_machine, dst_machine, dst_ip, dst_interface, output_file, dst_port_min=2000, dst_port_max=2999, payload_size=10, step=1) → dict`

Generate a synthetic TCP traffic capture for pcap analysis exercises. Uses 3 consecutive steps starting at `step`. The pcap file is written to `output_file` inside `dst_machine` (must be under `/shared/` for host-side analysis). Returns a dict with the captured port numbers and one TCP frame chosen from each direction:

Key	Description
`server_port`, `client_port`	TCP port numbers used
`packet_src_to_dst`	1-based frame number (src → dst direction)
`packet_src_to_dst_tcp_window`	TCP window field of that frame
`packet_src_to_dst_absolute_seq_number`	Absolute sequence number
`packet_src_to_dst_absolute_ack_number`	Absolute acknowledgment number
`packet_dst_to_src`	Same fields for the dst → src direction

The dict is populated at `step+2` via a host callback after the capture finishes.

`setup_tcp_client_server(net_scheme, src_machine, dst_machine, src_ip, dst_ip, secret, dst_port_min=3000, dst_port_max=3999, interval=3, step=1) → dict`

Deploy a persistent background TCP client/server pair for generating live traffic throughout the lab. Uses 2 steps. The server runs on `dst_machine`, the client on `src_machine`; both survive for the lifetime of the lab. Returns `{'server_port': int, 'client_port': int}`.

Used in `Grade.grade()` (evaluation):

```
from pcap_gen import check_zero_window_probe, get_frame_info
```

`check_zero_window_probe(grade, file, max_length, packet_number) → bool`

Return `True` if the packet at 1-based `packet_number` in the pcap file (relative to the project shared directory) is a valid TCP Zero Window Probe. Checks: file within `max_length` KiB; target packet has ≤ 1 byte payload; a prior packet in the reverse direction advertised window=0; target SEQ equals `SND.UNA` or `SND.UNA-1`.

`get_frame_info(grade, filename, max_length, frame_number) → dict | None`

Return all parsed fields for a single frame from a pcap file (relative to the project shared directory). Returns `None` if the file is missing, too large, not a valid pcap, or `frame_number` does not exist. The returned dict always contains `frame_number`, `frame_link_type`, `captured_length`, `original_length`, `timestamp_sec`, `timestamp_usec`; plus protocol-specific fields for Ethernet, Linux cooked capture, ARP, IPv4, IPv6, ICMP, TCP, and UDP headers.

3.10 Module-level Attributes

Declare these at the top level of `srelab.py` to customize behavior. They are read with `getattr(module, name, default)`, so any attribute may simply be omitted to keep the default.

3.10.1 Lab identity and presentation

Attribute	Type	Default	Description
`title`	`str | Transl`	filename without `.py`	Lab title shown in the GUI tab header and in exported reports. Pass a `TranslatedText` (via `tr()` / `make_tr()`) for multilingual titles. The same title is also surfaced in the GUI's *Open project* picker once an admin has run `sre make-titles` to materialise per-directory `titles.json` sidecars (see `sre make-titles` in the CLI reference)
`defau`	`str`	`'en'`	Language code used to resolve `TranslatedText` values when the GUI's locale is unavailable and when `make_tr()` is called without an explicit `default_language=`

3.10.2 Student self-evaluation

Attribute	Type	Default	Description
allow_self_	bool	Fals	If True, students may trigger their own evaluation from the GUI
delay_betwe	int	0	Cooldown in seconds between two student self-evaluations (only enforced for sre --user eval --auto-eval, i.e. when the student presses **Start evaluation** in the GUI; periodic background evals do not count). The remaining delay is stored under params.self_grade_timestamp_dir (/var/lib/sre/last_self_grades/{lab_name})
no_mark_on_	bool	Fals	During a student-triggered evaluation, hide numeric grades and show only the OK / MEH / FAIL letters
hide_potent	bool	Fals	During a student-triggered evaluation, drop grade-list entries whose grade and max_grade are both 0 (i.e. penalty rubrics that have not yet fired). Has no effect on instructor-triggered evals

3.10.3 Automatic evaluation

Attribute	Type	Default	Description
eval_interval_	int	0	Auto-eval interval in seconds when no exam is configured. 0 disables periodic evaluation
eval_before_ex	bool	False	If True, run a final evaluation when the student closes the project
use_numerical_	bool	params.use_numerical_marks_by_ (True)	If False, the lab reports OK / MEH / FAIL letters instead of numeric scores
display_marks_	bool	params.display_marks_in_auto_e (False)	If False, the numeric mark is hidden from students during automatic (periodic) evaluations even when use_numerical_marks=True
maximum_mark	int | float	params.default_maximum_mark (20)	Upper bound used when normalizing the lab's total grade

3.10.4 Archives and recordings

Attribute	Type	Default	Description
archive_di	list	[]	Extra directories to write evaluation archives (.zst) to, in addition to params.archive_dirs
files_to_s	list	[]	Container paths whose contents are copied into each evaluation archive (under the files key). Useful for capturing config files, logs, or pcap traces produced by the student
record_ses	bool | None	None	Override whether terminal sessions opened from the GUI are recorded (asciinema). None defers to the global / exam setting
save_recor	list	[]	Extra directories to write session-record archives (.tar.gz) to during an exam, in addition to params.save_records_dir (/var/lib/sre/archives). Read by sre save-records and by the automatic save loop during eval-exam
save_recor	int	params.default_save_rec (60)	Minimum delay in seconds between two automatic session-record saves for the same project during an exam. Set to 0 to disable automatic saving for this lab

3.10.5 State machine and flavors

Attribute	Type	Default	Description
allow_user_	bool	Fals	If True, students can apply states decorated with @sre_state(user_allowed=True) from the GUI. States decorated with user_allowed=False remain instructor-only
flavor_form	bool	Fals	If True and a Flavor subclass is defined, the GUI shows the flavor form before calling sre start, letting the student parameterize the lab

3.10.6 Filesystem and export

Attribute	Type	Default	Description
shared_path	bool	Fals	Create /home/sre/{running_lab_name}/ (mode 0o777) and bind-mount it into each container for file exchange between host and containers
export_kathar	bool	True	If False, disable **File → Export** in the GUI and refuse sre export for this lab

3.10.7 Schema rendering

These attributes tweak the graphviz topology diagram drawn for the GUI's *Schema* tab and embedded in the PDF produced by sre export.

Attribute	Type	Default	Description
show_nat_netwc	bool	params.default_show_nat_netw (True)	If False, hide the host-network vertex even when machines are marked bridged=True
host_network_r	str	params.default_host_network_ ("Internet")	Label shown on the host-network vertex
host_network_c	str	params.default_host_network_ ("deepskyblue")	Fill color of the host-network vertex (graphviz color name or #rrggbb)
host_network_e	bool	params.default_host_network_ (False)	If True, draw one separate host-network vertex per bridged=True machine instead of a single shared one
host_network_e	floa	params.default_host_network_ (1.0)	Relative length (graphviz len) applied to edges between bridged=True machines and the host vertex. < 1 shortens them, > 1 lengthens them
schema_splines	str	params.graphviz_default_spli ("curved")	Graphviz splines attribute (e.g. "line", "ortho", "curved", "polyline")
schema_overlap	str	params.graphviz_default_over ("prism")	Graphviz overlap attribute controlling how the layout resolves node overlaps

3.11 Complete Minimal Example

```
# /opt/sre/lab/example/srelab.py

from dataclasses import dataclass
from ipaddress import IPv4Interface, IPv4Network
from typing import Dict

from SRE.lib_sre import Data0, NetScheme0, Grade0, sre_state, make_tr
from ips import random_ipv4networks, random_ipv4s
from net_config import set_net_config_entry, NetConfigEntry,␣
↪get_net_config_from_topology, set_ip_forward
from state_helpers import set_nat_gateway

tr = make_tr('en')
title = tr("Example: Router", fr="Exemple : Routeur")
allow_self_grade = True
delay_between_self_grade = 60
eval_interval_without_exam_mode = 120


@dataclass(slots=True)
class Data(Data0):
    @classmethod
    def generate(cls, flavor=None):
        d = cls()
        d.nets.lan1, d.nets.lan2 = random_ipv4networks([24, 24],␣
```

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```
↪from_private_network=True)
        ips1 = random_ipv4s(d.nets.lan1, 2)
        ips2 = random_ipv4s(d.nets.lan2, 2)
        d.ips.router_lan1 = IPv4Interface(f'{ips1[0]}/{d.nets.lan1.prefixlen}')
        d.ips.client1 = IPv4Interface(f'{ips1[1]}/{d.nets.lan1.prefixlen}')
        d.ips.router_lan2 = IPv4Interface(f'{ips2[0]}/{d.nets.lan2.prefixlen}')
        d.ips.client2 = IPv4Interface(f'{ips2[1]}/{d.nets.lan2.prefixlen}')
        return d


class NetScheme(NetScheme0):
    _machine_specs = {
        'router': {'color': 'green'},
        'client1': {},
        'client2': {},
    }
    _network_specs = {
        'lan1': {'color': 'yellow'},
        'lan2': {'color': 'cyan'},
    }
    _topology = {
        'lan1': ['router', 'client1'],
        'lan2': ['router', 'client2'],
    }

    def __init__(self, data, running_lab_name):
        super().__init__(data=data, running_lab_name=running_lab_name)
        self.informations = tr(
            "## Router Lab\nConfigure routing between lan1 and lan2.",
            fr="## TP Routage\nConfigurez le routage entre lan1 et lan2.",
        )
        d = data
        self.net_config = get_net_config_from_topology(net_scheme=self, gateway="router")
        # self.net_config: Dict[str, NetConfigEntry] = {
        #     'router': [([d.ips.router_lan1], [(d.nets.lan2, d.ips.router_lan2)]),
        #                ([d.ips.router_lan2], [])],
        #     'client1': [([d.ips.client1], [(IPv4Network('0.0.0.0/0'),␣
↪d.ips.router_lan1)])],
        #     'client2': [([d.ips.client2], [(IPv4Network('0.0.0.0/0'),␣
↪d.ips.router_lan2)])],
        # }

    @sre_state(user_allowed=False)
    def initial(self):
        for machine, config in self.net_config.items():
            set_net_config_entry(net_scheme=self, machine_name=machine, nc_entry=config)
        self.cmd('router', 'sysctl -w net.ipv4.ip_forward=1')
        for m in self.get_visible_machine_names():
            set_ip_forward(net_scheme=self, machine_name=m, ip_forward=(m == "router"))
        set_nat_gateway(net_scheme=self, machine="router")
```

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```
class Grade(Grade0):
    def grade(self):
        super().grade()
        d = self.data

        # Test cross-network reachability
        result, code = self.test('client1', f'ping -c1 -W2 {d.ips.router_lan2.ip}',␣
↪timeout=5)
        self.add_grade_element('Cross-network ping', max_grade=10)
        if code == 0:
            self.set_grade('Cross-network ping', 10)

        # Ask a question
        answer = self.question_text(
            title=tr('Routing protocol', fr='Protocole de routage'),
            description=tr('Which mechanism routes packets between the two LANs?',
                           fr='Quel mécanisme route les paquets entre les deux LANs ?'),
        )
        self.add_grade_element('Routing answer', max_grade=5)
        if 'ip_forward' in answer or 'forward' in answer.lower():
            self.set_grade('Routing answer', 5)
```

CHAPTER

FOUR

TRANSLATING A LAB

4.1 Internationalization with `tr()` / `make_tr()`

```
from SRE.lib_sre import make_tr

default_language = 'en'
tr = make_tr(default_language)

title = tr("SSH Lab", fr="TP SSH")
# In NetScheme.__init__:
self.informations = tr("## Description\nConfigure SSH.",
                       fr="## Description\nConfigurez SSH.")
```

`tr(default, **lang_overrides)` returns a `TranslatedText` dict subclass. Call `.resolve('fr')` to get the string for a given language.

`make_tr(default_lang, translations=None)` returns a `tr()` function bound to *default_lang* as the key for the first positional argument. When `translations=` is omitted, the caller module's globals are inspected for a `_TRANSLATIONS` dict at each call; pass it explicitly to avoid relying on module-level state.

4.2 Marking strings that must NOT be translated: `no_tr()`

```
from SRE.lib_sre import no_tr

label = no_tr("MTU")              # short internal label, identical in every language
```

`no_tr()` is an identity passthrough at runtime — it returns the string unchanged. Its sole purpose is to flag a literal so the translation toolchain leaves it alone: it is never wrapped in `tr()` and never registered in `_TRANSLATIONS`. Use it for short internal labels (grade-element titles, protocol names, command names) that are not natural-language prose.

By default, `prepare-sre-translations` already wraps any whitespace-free single-token bare string in `no_tr()` rather than `tr()` (see `--translate-isolated-words` below to override).

4.3 Centralizing translations with `_TRANSLATIONS`

For labs with many strings, keeping translations inline in every `tr()` call becomes unwieldy. The `_TRANSLATIONS` dict centralises them all in one place:

```
from SRE.lib_sre import make_tr

default_language = 'en'
tr = make_tr(default_language)

_TRANSLATIONS = {
    'fr': {
        "SSH Lab":                  "TP SSH",
        "Configure the SSH server": "Configurez le serveur SSH.",
        "SSH port":                 None,   # not yet translated
    },
}

title       = tr("SSH Lab")
description = tr("Configure the SSH server")
```

`None` marks a string that still needs translation; `tr()` falls back to the default-language text in that case.

You can pass the dict explicitly to `make_tr()`, which avoids any reliance on module-level state:

```
tr = make_tr('en', translations=_TRANSLATIONS)
```

4.3.1 Where to put _TRANSLATIONS

Two valid placements; the choice affects which `tr()` calls can resolve translations at import time.

At the top of the file (default for `prepare-sre-translations`): `_TRANSLATIONS` is defined right after `tr = make_tr(...)`, before any `tr()` call. Every `tr()` call — including module-level and class-body ones evaluated at import time — can look up its translation. Inline `fr=`/`de=`/… kwargs are not needed.

At the end of the file (`prepare-sre-translations --translations-at-the-end`): `_TRANSLATIONS` lives at the very bottom, out of the way of the lab code. Because Python evaluates the module top-to-bottom, any `tr()` call in the module body or in class bodies runs *before* `_TRANSLATIONS` exists. To keep those import-time calls correct, `prepare-sre-translations` keeps (and re-adds on each run) inline lang kwargs on every such call. `tr()` calls inside method bodies — which only run at lab-execution time — do not need the kwargs and remain bare.

4.3.2 Dynamic strings with format placeholders

For strings that embed runtime values, use `{placeholder}` syntax in `_TRANSLATIONS` values and call `.format()` on the result:

```
_TRANSLATIONS = {
    'fr': {
        "The machine {machine} is configured": "La machine {machine} est configurée",
    },
}

msg = tr("The machine {machine} is configured").format(machine=m)
```

The static key `"The machine {machine} is configured"` is what `tr()` looks up in `_TRANSLATIONS`; `.format()` applies `str.format` to every language value at once and returns a new `TranslatedText`.

This is the equivalent of the inline f-string form:

```
msg = tr(f"The machine {m} is configured",
         fr=f"La machine {m} est configurée")
```

The inline f-string form still works and is fine for one-off strings. Use `_TRANSLATIONS` + `.format()` when the same template appears in many places, or when you want the translation toolchain (`check-sre-translations`, `add-sre-translations`) to see the string as a static literal.

4.4 Translation toolchain

Three command-line tools manage the complete translation lifecycle.

4.4.1 1. `prepare-sre-translations` — migrate and scaffold

`sbin/prepare-sre-translations` reads a lab file and:

- wraps bare translatable strings (module-level `title`, `description`, `informations`, `lab_name`; `self.informations` / `self.description`; `title=`/`description=`/`informations=` kwargs; first positional arg of `question_text`/`question_form`/`question_dummy`/`add_grade_element`/`add_grade_part`) in `tr()` — or in `no_tr()` if the string is a single whitespace-free token
- recurses into `BinOp` (`"x" + var`) and `IfExp` (`"x" if c else "y"`) expressions in those positions, wrapping each leaf
- lowers bare f-strings to `tr("template").format(var=var, ...)`
- creates or updates `_TRANSLATIONS` with one entry per string per language (value `None` for strings not yet translated)
- inserts `tr = make_tr(...)` and the matching `from SRE.lib_sre import make_tr` (and `no_tr` if needed) if absent

```
prepare-sre-translations [--move-tr-strings] [--default-language xx]
                         [--change-default-language xx]
                         [--translations-at-the-end]
                         [--translate-isolated-words] file
```

Option	Effect
(no options)	Wrap bare strings; create/update `_TRANSLATIONS` after `tr = make_tr(...)`; infer default language from file or assume `en`
`--default-xx`	Declare (or confirm) the default language; error if the file already uses a different one
`--change-d xx`	Pivot the file's source language to `xx`: rewrite every `tr()` literal to its `xx` translation, re-key `_TRANSLATIONS` so the inner keys are the new source texts, and preserve the previous default as a regular language. Errors out if any `tr()` string lacks an `xx` translation, or any translatable string is still bare.
`--move-tr-`	Strip inline `fr=`/`de=`/… kwargs from existing `tr()` calls and move them into `_TRANSLATIONS`
`--translat`	Place `_TRANSLATIONS` at the very end of the file; keep/add inline lang kwargs on every import-time `tr()` call (module/class body) so they don't depend on the late definition
`--translat`	Also wrap whitespace-free single-token strings in `tr()` (default: wrap them in `no_tr()`)

`--default-language` and `--change-default-language` are mutually exclusive.

Strings already wrapped in `no_tr(...)` are left untouched: never wrapped in `tr()` and never added to `_TRANSLATIONS`.

Typical first run on an existing lab:

```
# Wrap bare strings, declare the language, migrate inline kwargs:
sbin/prepare-sre-translations --default-language en --move-tr-strings lab/my_lab.py
```

The file is modified in-place. Run it repeatedly — it is idempotent: already-wrapped strings and existing translations are left unchanged.

4.4.2 2. `check-sre-translations` — verify consistency

`sbin/check-sre-translations` performs a static (AST-level) analysis and reports three categories of problem:

Category	Meaning
`MISSING`	String appears in a `tr()` call but has no entry in `_TRANSLATIONS`
`UNTRANSLATED`	Entry exists but its value is `None` (translation not yet done)
`VANISHED`	Entry exists in `_TRANSLATIONS` but no `tr()` call uses that string any more

```
sbin/check-sre-translations lab/my_lab.py
# lab/my_lab.py: MISSING       [fr] 'SSH port'
# lab/my_lab.py: UNTRANSLATED  [fr] 'Configure the SSH server'
# lab/my_lab.py: VANISHED      [de] 'Old string'
# lab/my_lab.py: ok (12 strings, ['fr', 'de'] languages)
```

Exit code is 0 if no issues are found, 1 otherwise. Suitable for CI.

Multiple files are accepted; let the shell expand globs:

```
sbin/check-sre-translations lab/s4/*.py
```

4.4.3 3. `add-sre-translations` — machine-translate missing strings

`sbin/add-sre-translations` calls an online translation service to fill in every `None` value for one target language, then writes the result back into the file.

```
add-sre-translations --language XX [--service YY]
    [--api-key KEY] [--credentials FILE] [--region REGION]
    [--libre-url URL] [--auto] file
```

Services (`--service`):

Service	Default credentials
`deepl` *(default)*	`--api-key` or `$DEEPL_API_KEY`
`google`	`--api-key` or `$GOOGLE_API_KEY` (REST); or `--credentials service-account.json` / `$GOOGLE_APPLICATION_CREDENTIALS`
`libre`	`--api-key` or `$LIBRE_API_KEY` (optional); `--libre-url` for a self-hosted instance (default: `https://libretranslate.com`)
`azure`	`--api-key` or `$AZURE_TRANSLATOR_KEY`; `--region` (default: `global`)
`amazon`	Standard AWS credential chain (`~/.aws/` or env vars); `--region` or `$AWS_DEFAULT_REGION`

Interactive mode (default when stdin is a TTY):

For each string the tool shows the machine translation and prompts:

```
[fr] 'SSH port'
     → 'Port SSH'
  Accept [Enter] or type correction (q to quit):
```

Press Enter to accept, type a correction, or `q` to stop early. Progress is always saved, even on Ctrl-C or an API error. Pass `--auto` to accept all suggestions without prompting.

Typical session:

```
export DEEPL_API_KEY=your-key-here

# Fill in French translations interactively:
sbin/add-sre-translations --language fr lab/my_lab.py

# Fill in German translations automatically:
sbin/add-sre-translations --language de --auto lab/my_lab.py

# Verify everything is done:
sbin/check-sre-translations lab/my_lab.py
```

4.5 Complete translation workflow

For a new lab starting from scratch in English:

```
# 1. Scaffold: wrap strings, create _TRANSLATIONS skeleton at the end of the file
sbin/prepare-sre-translations --default-language en --translations-at-the-end␣
↪lab/my_lab.py

# 2. Verify what needs translating
sbin/check-sre-translations lab/my_lab.py

# 3. Fill in French with interactive review
sbin/add-sre-translations --language fr lab/my_lab.py

# 4. Final check - should report no issues
sbin/check-sre-translations lab/my_lab.py
```

For a lab that already uses inline `tr("text", fr="traduction")` kwargs:

```
# Migrate inline kwargs into _TRANSLATIONS in one step
sbin/prepare-sre-translations --move-tr-strings lab/my_lab.py

# Review and fill in anything still missing
sbin/check-sre-translations lab/my_lab.py
sbin/add-sre-translations --language fr lab/my_lab.py
```

4.6 Switching the default language: `--change-default-language`

`prepare-sre-translations --change-default-language xx` pivots the file's source language to `xx`. The resulting file is *semantically equivalent* to the original — same translations, just keyed off a different source — and `check-sre-translations` should report it clean immediately after the pivot.

It performs four edits:

- `default_language = '...'` is set to `xx`;
- the literal first argument of `make_tr('...')` is set to `xx` (a `Name` arg like `make_tr(default_language)` is left as-is and inherits the new value);
- every `tr("old_text", ...)` call's first positional arg is replaced with its `xx` translation; the `xx=...` kwarg, if present, is dropped (it is now the default), and if the call had any inline language kwargs, the previous default is added as `<prev>="old_text"` to preserve the inline-kwarg style;
- `_TRANSLATIONS` is re-keyed: the inner keys flip from the previous source texts to the new `xx` source texts, the `xx` top-level entry disappears, and the previous default appears as a regular language entry mapping new-source → previous-source text.

4.6.1 Prerequisites

The pivot only runs if the file is already fully prepared and every `tr()` literal has an `xx` translation reachable via inline kwargs or `_TRANSLATIONS`. The script reports an error and leaves the file untouched when it finds any of:

- a `tr()` literal without an `xx` translation (`add-sre-translations --language xx` is the suggested fix);
- a `tr()` call whose first argument is not a string literal (variables, f-strings, expressions cannot be statically rewritten — lower or inline these first by running `prepare-sre-translations` without `--change-default-language`);
- a bare translatable string still waiting to be wrapped (same fix: run `prepare-sre-translations` without `--change-default-language` first).

`--change-default-language` is incompatible with `--default-language`, and rejects a pivot to the language that is already the file's default.

4.6.2 Example: switching to English as a pivot before translating

Machine-translation services produce noticeably better results when English is the source language, especially for less-common target pairs. If your lab is currently in another language but you want to pivot to English so every subsequent `add-sre-translations --language fr/de/es/...` call uses English as `source_lang`:

```
# 1. Fill in English translations first - these become the new source.
sbin/add-sre-translations --language en lab/my_lab.py

# 2. Pivot the file: tr() literals + _TRANSLATIONS keys + declarations all move to en.
sbin/prepare-sre-translations --change-default-language en lab/my_lab.py

# 3. Every add-sre-translations call now uses English as source_lang.
sbin/add-sre-translations --language fr lab/my_lab.py
sbin/add-sre-translations --language de lab/my_lab.py
sbin/add-sre-translations --language es lab/my_lab.py
```

CHAPTER

FIVE

INSTALLATION AND SETUP

5.1 Prerequisites

- A Linux host. `scripts/install.sh` knows how to `apt-get install` missing packages on Debian and Ubuntu; on anything else you must install the prerequisites yourself.
- **Root** access. The installer refuses to start unless `EUID == 0` (run with `sudo`).
- The **Docker daemon running**, with `/var/run/docker.sock` present. The installer reads the socket's GID and grants the `sre` user access to it; if the socket is missing it aborts with a hint to start the daemon.
- **Python 3.13**. On Ubuntu < 24.04 you may need the `deadsnakes` PPA; on Debian < 13 you need to upgrade or build it from source. The installer will offer to `apt-get install python3.13` on Debian-likes.

These tools must already be on `PATH` (always present on a POSIX system): `sed grep stat getent useradd groupadd usermod install`.

These tools are auto-installed via `apt-get` on Debian-likes if missing:

- `docker` (any variety) `asciinema make gcc graphviz python3.13`

On other distributions install them with your package manager before running the installer.

5.2 Deployment layout

SRE's runtime root defaults to `/opt/sre`, but the installer makes it configurable. Clone the repo (`git clone sysreseval/sysreseval`) or untar the release into `/opt/sre` — or into any other location you prefer.

`scripts/install.sh` detects where it is being run from and offers that path as the default — `/opt/sre` when run from `/opt/sre/scripts/`, otherwise the parent of the script directory. The chosen value is patched into `src/SRE/params.py` (`main_sre_dir`) and substituted for `/opt/sre` in the sudoers rule, the `.desktop` file, and the `sre-preload-images.service` unit at install time.

The installer is **interactive** — it asks roughly ten configuration questions, then creates the `sre` user, installs a sudoers file, and runs the build — and is meant to be run **once per build**, not once per student workstation. The trick to scaling from a single interactive run to a classroom of student machines is to share the resulting runtime tree (and replay a small per-host setup on each workstation). The rest of this document uses `/opt/sre` as the canonical example path; substitute your `main_sre_dir` wherever it appears if you chose something else.

Two common topologies:

Single read-only NFS export (recommended for classrooms). Clone the repo on a build server inside the directory you intend to export, run `sudo ./scripts/install.sh` there once, then export the

directory read-only and mount it as `/opt/sre` on every student workstation. Updates take effect everywhere as soon as the export is refreshed, and students cannot tamper with the binaries or with the project files. The shell wrappers in `sbin/` and `bin/` don't write to their own tree, and all runtime state lives elsewhere — `/var/lib/sre` (`sre_pub_dir`) and `/home/sre` (`sre_user_public_dir`) stay on the local workstation — so a read-only mount works.

Build once, replicate per workstation. Run the installer once on a reference machine cloned at `/opt/sre`, then `rsync` (or image-clone) the resulting `/opt/sre` tree to every workstation. Running `install.sh` interactively on each machine is impractical for more than a handful of hosts.

Either way, each student workstation still needs a few per-host bits that aren't part of the `/opt/sre` tree:

- the `sre` system user/group, with the UID/GID baked into `params.py` at build time
- `/etc/sudoers.d/sre`
- membership of `sre` in the local `docker` group
- a copy of `scripts/etc/sre_bash_completion` into `/etc/bash_completion.d/sre`
- a copy of `scripts/etc/sysreseval.desktop` into `/usr/share/applications/`
- raised inotify limits and other post-install steps (see *Post-install steps*)
- for the NFS topology, the mount itself

Script these from the relevant sections of `scripts/install.sh` if you have many machines.

The `/opt/sre` tree must contain `lab/`, `lib/`, `graphics/`, `translations/`, `locale/`, `bin/`, `sbin/`, `src/`, `venv/`. Lab authors also browse `src/` while writing `srelab.py` files. See *Runtime & Internals* for the expected layout.

If you'd rather build in a separate working checkout (CI runner, personal dev tree) and move the result over afterwards, `rsync` or bind-mount the built tree into `/opt/sre` once `make install` has finished, then continue with the *Post-install steps*.

5.3 Running the installer

```
sudo ./scripts/install.sh
```

The script is interactive. It asks, in order:

Prompt	Default	Purpose
`main_sre_`	`/opt/sre` if the script is at `/opt/sre/scripts/`, otherwise the script's parent directory	Runtime root for SRE binaries, libs, labs, graphics, translations, locale. Substituted for `/opt/sre` in the sudoers rule, `.desktop` file, and systemd unit at install time.
`sre` UID	`1100`	UID of the system user that owns running labs.
`sre` GID	`1100`	GID of the matching group.
`sre_pub_d`	`/var/lib/sre`	Public state: running projects, archives, `exam.json`.
`sre_user_`	`/home/sre`	Per-lab shared `/home/sre/{running_lab}` mounts.
`allow_pri`	`y`	Whether `srelab.py` files may declare privileged containers.
`execute_c`	`shell`	`shell`, `split`, or `False` — controls how lab `host_cmd()` calls are executed.
extra authorized src dirs	`/home`	Extra directories (beyond `main_sre_dir + '/lab'`, which is always included) from which `srelab.py` files may be loaded. Type `-` for none to restrict loading to the lab directory only.
Admin UIDs / GIDs	*(none)*	Extra UIDs/GIDs that SRE treats as administrators.

The Docker socket GID is detected automatically (no prompt).

After you confirm the summary, the installer:

1. Saves a backup of `params.py` to `params.py.bak.<YYYYmmdd-HHMMSS>`.
2. Patches the answers into `src/SRE/params.py` (single-line `key = value` substitutions, each verified after the edit).
3. Runs `make remove-debug-mode` if `debug_mode = True` is currently set — `make install` refuses to build while debug mode is on.
4. Creates the `sre` system group and user (`nologin` shell, home `/home/sre`) if they don't already exist. If a user/group with the requested name exists with a different UID/GID, it is **kept as-is** rather than overwritten — the installer warns and continues.
5. Adds `sre` to the Docker group resolved from the socket GID.
6. Installs `/etc/sudoers.d/sre` (**required** — the `sysreseval` GUI shells out to `sudo /opt/sre/sbin/sre --user` via `sre-wrapper`, so labs cannot start without this rule), validated by `visudo -c` before being kept. The rule it installs is:

   ```
   Defaults!/opt/sre/sbin/sre env_keep += "USER_USERNAME SRE_XAUTH_COOKIE"
   ALL  ALL= NOPASSWD: /opt/sre/sbin/sre --user *
   ```

   The rule grants passwordless access to every user on the host, not a specific group — `sre-wrapper` is the only entry point and it always passes `--user` with the caller's real login, so widening the sudoers scope doesn't lower the security envelope. Restrict it to a specific group (e.g. `%etudiant`) if your site policy requires it.
7. Optionally installs `scripts/etc/sysreseval.desktop` to `/usr/share/applications/` so the GUI appears in the desktop application menu.
8. Optionally installs `scripts/etc/sre_bash_completion` to `/etc/bash_completion.d/sre` so the `sre` CLI gets bash completion system-wide.
9. Optionally installs the `sre-preload-images.service` systemd unit (oneshot, requires `opt-sre.mount`; not enabled by default — enable with `systemctl enable --now sre-preload-images.service` once

`/opt/sre` is mounted).

10. Runs `make venv` then `make install` (which is `check-debug-mode` + `sre-wrapper` + `wrappers`).
11. Optionally creates symlinks in `/usr/local/bin/` and `/usr/local/sbin/` pointing to each executable under `main_sre_dir/bin/` and `main_sre_dir/sbin/`, so `sre`, `sysreseval`, and `sre-wrapper` can be launched without their full path. Existing non-symlink files at the same paths are left untouched (with a warning).

5.4 Install manually

If you prefer to skip `scripts/install.sh` (e.g. on a non-Debian distribution, or to integrate with your own configuration management), perform the same steps by hand from a checkout of the repo:

1. **Create the `sre` system group and user.** Pick a UID/GID (the installer defaults to `1100`). The user needs a real home for shared `/home/sre/{running_lab}` mounts but no shell login:

```
groupadd --system --gid 1100 sre
useradd --system --uid 1100 --gid 1100 --home-dir /home/sre --shell /
↪usr/sbin/nologin sre
install -d -o sre -g sre -m 0755 /home/sre
```

2. **Add `sre` to the Docker group** so it can talk to `/var/run/docker.sock`:

```
usermod -aG "$(stat -c %G /var/run/docker.sock)" sre
```

3. **Edit `src/SRE/params.py`** to match your site. At minimum review:
   - `main_sre_dir` — runtime root for SRE binaries, libs, labs, graphics, translations, locale (default `/opt/sre`). If you change this, also rewrite the literal `/opt/sre` in the sudoers rule, `.desktop` file, and `sre-preload-images.service` snippets below.
   - `sre_uid`, `sre_gid` — must match the user/group created above.
   - `docker_gid` — GID of the group owning `/var/run/docker.sock`.
   - `sre_pub_dir` — public state directory (default `/var/lib/sre`).
   - `sre_user_public_dir` — per-lab shared directory (default `/home/sre`).
   - `allow_privileged_machines` — whether `srelab.py` files may declare privileged containers.
   - `execute_commands_on_host` — `'shell'`, `'split'`, or `False`.
   - `authorized_src_dir` — list of directories from which `srelab.py` files may be loaded. Must include `main_sre_dir + '/lab'`; defaults to also including `/home`.
   - `admin_uids`, `admin_gids` — extra UIDs/GIDs treated as administrators.
   - `debug_mode` — **must be `False`** before building (`make wrappers` refuses otherwise). Use `make remove-debug-mode` if needed.
4. **Install the sudoers rule** — required for `sysreseval` to start labs (the GUI invokes `sudo /opt/sre/sbin/sre --user` through `sre-wrapper`). The rule allows every user on the host to invoke the wrapper without a password; restrict the first column to a specific group (e.g. `%etudiant`) if your site policy requires it. Then validate before keeping the file:

```
Defaults!/opt/sre/sbin/sre env_keep += "USER_USERNAME SRE_XAUTH_COOKIE"
ALL  ALL= NOPASSWD: /opt/sre/sbin/sre --user *
```

```
visudo -cf /etc/sudoers.d/sre && chmod 0440 /etc/sudoers.d/sre
```

5. **(Optional) Install the desktop entry** so the GUI appears in the desktop application menu:

```
install -m 0644 -o root -g root scripts/etc/sysreseval.desktop /
↪usr/share/applications/sysreseval.desktop
```

6. **(Optional) Install the image pre-pull systemd unit** from `scripts/etc/sre-preload-images.service` into `/etc/systemd/system/`. It requires `opt-sre.mount` and is not enabled by default — see *Pre-loading Docker images* below.

7. **(Optional) Install bash completion** for the `sre` CLI. The script under `scripts/etc/sre_bash_completion` completes subcommands, running lab names (read from `/var/lib/sre/projects/`), available labs (via `sre list`), and option arguments. Install it system-wide:

```
cp scripts/etc/sre_bash_completion /etc/bash_completion.d/sre
```

   Or source it per-user from `~/.bashrc`:

```
. /path/to/sre_bash_completion
```

8. **Build the venv and the binaries:**

```
make venv      # creates venv/ with Python 3.13 deps - see Dependencies below
make install   # check-debug-mode + sre-wrapper + wrappers
```

You can also run the CLI and GUI directly from source against the venv:

```
source venv/bin/activate
python3 -W ignore src/sre.py <command>     # CLI
python3 src/sysreseval.py                  # GUI
```

Most `sre` subcommands need root and a working Docker setup, but the pure-Python paths (e.g. `sre cat`, `sre sheet`, `sre outline`) work from a venv as a regular user against pre-existing archive files.

5.5 Post-install steps

Once the installer has built the binaries, finish the deployment:

5.5.1 1. Deploy and verify

1. Make sure the built tree is reachable at `/opt/sre` on every workstation that will run labs. If you ran the installer from `/opt/sre` directly (per *Deployment layout*) this is already done; otherwise copy/bind-mount the built tree into place, or expose it via the read-only NFS export and mount it as `/opt/sre` on each workstation.

2. Verify Docker access for the `sre` user:

```
sudo -u sre docker ps
```

   The Docker images SRE needs are pulled from Docker Hub on first lab start; no local image build is required.

3. Make sure `/opt/sre/bin` and `/opt/sre/sbin` are reachable on `PATH` so that `sysreseval`, `sre-wrapper`, and `sre` can be launched without their full path. The installer offers to create the symlinks under `/usr/local/bin` and `/usr/local/sbin` for you; if you skipped that prompt (or installed by hand),

either add the two directories to the system-wide `PATH` (e.g. via `/etc/profile.d/sre.sh`), or create the symlinks manually:

```
ln -s /opt/sre/bin/*  /usr/local/bin/
ln -s /opt/sre/sbin/* /usr/local/sbin/
```

5.5.2 2. Raise inotify limits

Privileged labs run `/sbin/init` (systemd PID 1) inside each container, and systemd reserves inotify watches per cgroup. Debian defaults (`max_user_instances = 128`) are quickly exhausted once two privileged labs run side by side, after which systemd in any new container exits at startup with `Failed to create control group inotify object: Too many open files`. Drop in the file shipped under `scripts/etc/`:

```
cp scripts/etc/sre-inotify.conf /etc/sysctl.d/60-sre-inotify.conf
sysctl --system
```

(Optional) Tune `src/SRE/params.py` further — see *Runtime & Internals*.

5.5.3 3. Restricting student access to Docker

For an exam to be meaningful, students must not be able to talk to Docker directly — otherwise a `docker exec -it ...` from a regular shell would let them connect to a forbidden container and bypass the lab's restrictions.

The recommended setup:

1. Remove students from the `docker` group entirely. Only `sre` (and any administrator account) should be a member.
2. Create a separate `docker-access` group and add the student accounts to it.
3. Toggle that group's access to `/var/run/docker.sock` with the two helpers shipped under `scripts/`:
   - `scripts/docker-allowed` grants `rw` on the socket to `docker-access` (run before non-exam sessions).
   - `scripts/docker-forbidden` revokes it (run before an exam).

Both helpers use `setfacl` on the Docker socket, so the change is immediate and survives until the next call.

5.5.4 4. Sharing evaluation archives for `sre watch`

`sre watch <directory>` runs an interactive terminal dashboard that monitors evaluation archives as they appear and alerts on student inactivity or errors. To use it during an exercise session or an exam, every workstation must drop its archives into a directory the instructor can read from one place. Set the `archive_dirs` attribute in `params.py` (or on some lab's `srelab.py`) so evaluations are written into the shared location in addition to the local one.

There are several ways to set up the shared hierarchy:

Single NFS share, server-side watch. Export one NFS directory, restrict read-write access to the `sre` user (e.g. `rw,no_root_squash` for the workstations, mounted only as `sre`), and run `sre watch` directly on the NFS server. This is the simplest setup and works well when the instructor's session is on the server itself.

Per-workstation NFS shares with nightly rotation. Export one subdirectory per student workstation, each writable only by that workstation. The script `scripts/misc/rotate-nfs-dirs.py` is meant to be invoked from `cron` (typically once a night). On each run it:

1. Renames `/home/sre-archives/current` to `/home/sre-archives/YYYY-MM-DD` (the previous day), so past-day archives are no longer reachable from any student computer.

2. Re-creates `/home/sre-archives/current/<machine>` for every workstation listed in the `MACHINES` constant.
3. Rewrites `/etc/exports.d/sre-archives.exports` so each subdirectory is exported only to the matching workstation IP.
4. `systemctl reload nfs-kernel-server`.

Edit the `MACHINES`, `BASE_DIR`, and `EXPORTS_FILE` constants at the top of the script to match your site before installing the cron job. `sre watch /home/sre-archives/current` then sees every workstation's live archives.

Read-only cross-mount for the instructor. In addition to either of the above, you can re-export the shared directory read-only to the instructor's workstation. The instructor can then run `sre watch` from their own machine without logging into the server.

5.5.5 5. Pre-loading Docker images

When an exam starts, every workstation pulls the same images at the same time — without precaution this can saturate the network and cause a meltdown. The fix is to pre-pull the images on each host ahead of time.

The unit `scripts/etc/sre-preload-images.service` (offered for installation by `scripts/install.sh`) runs once at boot and shells out to:

```
/opt/sre/sbin/sre preload-images --random-delay 120 /opt/sre/lab/
```

The `--random-delay` spreads the pulls over a 120-second window so workstations don't hit Docker Hub simultaneously. The unit requires `opt-sre.mount` (i.e. `/opt/sre` mounted as a systemd mount unit) and is **not enabled by default**:

```
systemctl enable --now sre-preload-images.service
```

5.6 Reference

5.6.1 Build targets

Target	Output
`make wrappers`	Marks `sbin/sre` and `bin/sysreseval` shell wrappers executable. Refuses to build while `params.debug_mode = True`.
`make sre-wrapper`	`bin/sre-wrapper` (small C helper that escalates via `sudo`).
`make install`	`check-debug-mode` + `sre-wrapper` + `wrappers`.
`make translations`	Compile Qt `.ts → .qm` and gettext `.po → .mo`.
`make docs`	API docs (pdoc) + main docs (Sphinx) → `docs/html/main/`, `docs/html/api/`, `docs/documentation.pdf`.
`make set-debug-mode` / `make remove-debug-mode`	Flip `debug_mode` in `src/SRE/params.py`.

`debug_mode` must be `False` for `make wrappers` and `make install`, but **must be `True`** to run exam-mode integration tests — the GUI then emits the JSON event stream those tests assert on.

5.6.2 Tests

```
make tests                          # unit tests (excludes test_exam_mode.py)
make test FILE=test_net_config.py   # single file
make functional-tests               # functional tests (test_functional.py)
make exam-tests                     # exam-mode integration tests (requires␣
↪debug_mode=True)
make all-tests                      # tests + functional-tests + exam-tests
```

Unit tests don't need Docker or Kathara — they cover serialization, topology, grading logic, and pure-Python helpers. Functional and exam-mode tests stand up more of the system: see `tests/conftest.py` for the shared fixtures (`mock_sre_args`, `tmp_lab_dir`, `tmp_pub_dir`).

Pytest runs with `-p no:cacheprovider` so the production install (`/opt/sre/`, read-only) doesn't try to write a cache.

5.6.3 Building the documentation

```
make docs            # api_doc + main_doc_pdf + main_doc_html
make main_doc_html   # Sphinx HTML only → docs/html/main/
make main_doc_pdf    # Sphinx LaTeX → docs/documentation.pdf
make api_doc         # pdoc API → docs/html/api/
```

The Sphinx targets pip-install `sphinx myst-parser furo` into the existing venv on demand; `api_doc` does the same for `pdoc`.

5.6.4 Dependencies

Installed by `make venv`:

Package	Purpose
`kathara`	Docker lab orchestration
`pyside6`	Qt6 GUI
`msgpack`	Efficient binary serialization
`zstandard`	Archive compression
`graphviz`	Network topology diagrams
`odfpy`	LibreOffice ODS spreadsheet export
`fpdf2`	PDF report generation
`markdown`	Lab description rendering
`netaddr`	MAC address handling
`pytest`	Test runner

Documentation builds also fetch `sphinx`, `myst-parser`, `furo`, and `pdoc` on demand.

CHAPTER

SIX

GUI REFERENCE — SYSRESEVAL

6.1 Launching

```
source venv/bin/activate && python3 src/sysreseval.py   # dev
sysreseval                                              # production
```

One instance per user. A PID file at `/tmp/sysreseval-{uid}.pid` kills any prior instance on startup.

6.2 Opening a project

File → Open Project (or the `+` tab) shows a dialog built from `sre list --with-titles`. Leaves display each lab's translated `title` (from per-directory `titles.json` sidecars, generated by `sre make-titles`), falling back to the filename without `.py`. If the lab's `Flavor` sets `flavor_form_at_startup = True`, a form runs first. A progress dialog then tracks image pulls and container starts (JSON on stderr from `sre start`); the project tab appears when every machine is up.

During an exam, **Open Project** is disabled — only pre-authorised exam labs open automatically.

6.3 Project tabs

Tab	Content
Schema	SVG topology rendered by graphviz at lab start. Scroll = zoom, click-drag = pan.
Informations	Markdown lab description from `self.informations` in `NetScheme.__init__()`.
Machines	Per-machine state (polled from Kathara every 1 s), NAT network, ports, plus a *Connect* button opening an external terminal.
Questions	Three types: `question_text` (free text), `question_form` (inline `@@{field:regex}@@` or `@@{field:>opt1|opt2}@@`), `question_dummy` (display only). Persisted to `answers/answers.json` on every edit.
Evaluation	Last grade table + *Start evaluation* button. Letter grades (OK/MEH/FAIL) appear when the lab sets `no_mark_on_self_grade`. A countdown replaces the button while `delay_between_self_grade` is active.
Terminals	Embedded terminals per machine, launched via `params.terminal_cmd_prefix`.

`answers/answers.json` also carries metadata (`hostname`, `login`, `fullname`, `email`, `language`, `answers_updated_at`, plus `exam_*` fields in exam mode); see *Archive format* for the full schema.

Start evaluation runs `sre eval --auto-eval` in a background thread. The periodic background eval (`eval_interval_without_exam_mode`) and `eval_before_exit` run plain `sre eval` (no cooldown, no log line, no stdout).

6.4 Exam mode

When `/var/lib/sre/exam.json` exists, the GUI switches modes: File → Open / Close / Close All are disabled, and answers re-save with an updated `exam_time_remaining` on every exam-config change.

Phas	Trigger	Shows / actions
Wait ing	`now < start_after`	SRE logo + countdown to `start_after`. Calls `sre pre-start-exam` once ~60 s before `start_after`.
Ac- tive	`start_after` ≤ `now < end` and exam projects up	Project tabs + remaining-time countdown. Calls `sre start-exam` once on entry, `sre eval-exam` every `eval_interval`; re-calls `pre-start-exam` if `labs` changes.
Ende	`now end_before`, or `now started_at + duration`	Calls `sre eval-exam` then `sre end-exam` once; displays the *That's All Folks* logo.

6.5 Settings

File → Settings (persisted in `~/.config/sysreseval`, INI):

Section	Setting	Default	Effect / applies
Interface	Font size	10 pt	Menus and dialog text; immediate
Terminal	Font size	12 pt	Terminals opened after the change
Terminal	Color scheme	Black on White	`black_on_white` / `white_on_black`; new terminals
Schema	Lines	Straight	Straight or curved edges; next lab open
Schema	Spacing	3 (0–9)	Node spacing; next lab open
Schema	Nodes	Shapes	Icons or geometric shapes; next lab open
Content	Font size	12 pt	Informations + Questions panes; immediate

CHAPTER

SEVEN

CLI REFERENCE — SRE

`sre --help` groups subcommands into five buckets: **dual** (admins + GUI via `sre-wrapper`), **admin-only**, **exam management**, **post-exam management**, and **internal** (GUI-only).

7.1 Global options

```
sre [--user] [--debug] <subcommand> ...
```

Option	Description
`--user`	Set by `sre-wrapper`; reads student login from `USER_USERNAME` (else `LOGNAME`). Mutually exclusive with `--debug`.
`--debug`	Verbose diagnostics to stderr.

Access is restricted to `root`, the `sre` user (`params.sre_uid = 1100`), members of `params.admin_uids`, or members of any `params.admin_gids` group. Non-privileged users may only run `cat`, `check-eval`, `re-eval`, `sheet`, and `outline`.

7.2 Dual commands

7.2.1 `sre start [-d data] [-p] [--flavor … | --flavor-json … | --set-flavor-name …] [--xauth-file <file>] <lab> [data_version]`

Starts a new lab instance, named `{timestamp}@@@{lab_name}@@@{username}`. Imports `srelab.py`, runs `Data.generate(flavor)` (or loads `-d data.json`), builds `NetScheme`, pulls images, deploys via Kathara, applies the `initial` state, and writes `info.json` + `scheme.svg`. See *Internals* for file-transfer and progress mechanics.

Argument / Option	Description
`lab`	Lab name (from `sre list`) or, with `-p`, a filesystem path.
`-p` / `--path`	Treat `lab` as a path.
`-d` / `--data`	Reuse an existing `data.json` instead of generating one.
`data_version`	Optional seed for `Data.generate()` (privileged only; incompatible with `-d`).
`--flavor key:val …`	Flavor field values (e.g. `--flavor nb:3 mode:hard`).
`--flavor-json <json>`	Flavor as JSON dict (used by the GUI).
`--set-flavor-name <name>`	Use a named preset `Flavor` (privileged only).
`--xauth-file <file>`	Read `SRE_XAUTH_COOKIE` from an X authority file instead of the env var (privileged only; for `x11_host=True` machines).

7.2.2 sre stop <running_lab>

Undeploys containers and removes the project directory from `/var/lib/sre/projects/`.

7.2.3 sre wipe

Stops every running Kathara lab and removes everything under `/var/lib/sre/projects/`.

7.2.4 sre connect [--shell <shell>] [--exec <argument…>] [--no-records] <running_lab> <device>

Opens an external terminal connected to a container.

Option	Description
`--shell <shell>`	Override the machine default shell (privileged only).
`--exec <argument…>`	Run one command via the shell and return (consumes rest of argv).
`--no-records`	Do not record the terminal session (privileged only).

7.2.5 sre eval [-p path] [--auto-eval] <running_lab>

Evaluates a running lab. The atomic lock file `.private/eval_in_progress` prevents concurrent evals.

`--auto-eval` marks a *user-triggered self-evaluation* (sent by the GUI's *Start evaluation*). With `--user`, it enables the `delay_between_self_grade` cooldown (outside exam mode), appends a line to `.private/auto_eval.log`, and prints the result JSON. Without `--auto-eval`, a `--user` invocation runs silently — no cooldown, no log, no stdout. The `auto_eval_count` (lines in `.private/auto_eval.log` before this run) is assigned to `self.auto_eval_count` and embedded in the archive's `answers`. If the lab sets `no_mark_on_self_grade`, letter grades (OK/MEH/FAIL) replace numeric grades. Results are archived as `.zst` — see *Archive format*.

7.2.6 sre state <running_lab> <state_name>

Applies `NetScheme.<state_name>()` to a running lab. File operations registered via `file()` / `append_to_file()` are injected via in-memory tar archives sent to `container.put_archive("/")`.

7.3 Admin-only commands

7.3.1 `sre exec [--shell <shell>] <running_lab> <device> <command…>`

Runs `shell -c <command>` in a container and forwards stdout/stderr/exit code. Default shell is `params.default_exec_shell`.

7.3.2 `sre eval-all [--display-grades / --no-display-grades]`

Evaluates every running lab instance concurrently. `--no-display-grades` suppresses the per-lab grade summary.

7.3.3 `sre check [-p] <lab> [<state>]`

Validates a lab module without deploying: imports it, runs `Data.generate()`, builds `NetScheme`, runs `initial()`, calls `Grade.grade()`. With `state`, also validates that state method.

7.3.4 `sre watch [--timeout <sec>] [--interval <sec>] <dir…>`

Live terminal dashboard scanning `<dir…>` for `.zst` archives. Default refresh: `params.default_dashboard_refresh_interval_in_watch_command`. `--timeout` is the inactivity-alert threshold (default 90 s).

Columns (from each archive):

Column	Source
HOSTNAME / LOGIN	`answers["hostname"]` / `answers["login"]`
LAB NAME	Middle segment of `running_lab_name`
GRADE	`total_grade / total_max`
ERR	Length of `errors`
LAST EVAL	`eval_date` (`YYYYmmddHHMMSS`)
TIME REMAINING	`answers["exam_time_remaining"]` minus elapsed since `answers["answers_updated_at"]`

Keys:

Key	Context	Action
`t`	Any	Toggle focus between Projects and Alerts
`↑` / `↓`	Projects / Alerts	Move selection
`P`	Projects	Dismiss selected project
`H`	Projects	Dismiss all projects from the selected hostname
`R`	Projects	Set a hostname regexp filter (empty = show all)
`U`	Either	Un-dismiss all
`d` / `Enter`	Alerts	Dismiss selected alert
`q` / `Ctrl-C`	Any	Quit
`?`	Any	Toggle help

Dismissals self-expire: inactivity alerts return on new archives, error alerts return when the count changes.

7.3.5 `sre preload-images [--random-delay <sec>] <file_or_dir…>`

Scans lab `.py` files for `image=` / `'image':` patterns and `docker pull`s each unique image. `--random-delay <sec>` waits 0–N seconds before pulling (smooths thundering-herd when invoked across many hosts).

7.3.6 `sre make-titles [-o <file> | -r] <directory>`

Generates `titles.json` sidecars so the GUI shows friendly labels (e.g. `static_routing.py` → "Static routing"). Labs without a `title` attribute are skipped.

Option	Description
`-o, --output-file <file>`	Write all titles to `<file>` (incompatible with `-r`).
`-r, --recursive`	One `titles.json` per directory containing labs (incompatible with `-o`).

Output keys match `sre list`: `"<name>.py"` for file labs, `"<dirname>"` for directory labs. Files are written atomically with sorted keys; stale entries are dropped at read time by `sre list --with-titles`.

```
{
  "static_routing.py": {"en": "Static routing", "fr": "Routage statique"},
  "tp_ssh":            {"en": "SSH lab",        "fr": "TP SSH"}
}
```

Typical usage: `sudo /opt/sre/sbin/sre make-titles -r /opt/sre/lab`.

7.4 Exam management

Exam state lives in `/var/lib/sre/exam.json` — see *Exam Reference* for the field schema. All field names are exposed as `params.exam_*` constants.

7.4.1 `sre set-exam [options]`

Creates or updates `exam.json`. Only provided options are updated; existing fields are preserved.

Option	Description
`--labs <lab[:flavor]…>`	Authorised labs, optionally with a named flavor preset (e.g. `tp1:hard`). Required on first creation.
`--start-after <datetime>`	Exam opens after this moment. ISO (`2026-06-01T09:00`) or time-only (`09:00`, today).
`--end-before <datetime>`	Exam closes before this moment.
`--duration <minutes>`	Max duration; ends at `started_at + duration` or `end_before`, whichever first. Effective only once formally started.
`--eval-interval <seconds>`	Period between automatic evals (default 60).
`--record-sessions <bool>`	Record terminal sessions (`true`/`false`/`0`/`no`).

7.4.2 `sre del-exam`

Removes `exam.json` and wipes all running projects.

7.4.3 `sre save-records -d <dir> [--only-last-record]`

Archives the `records/` directories of running projects as `.tar.gz` files in `<dir>`. `--only-last-record` deletes prior archives for the same project so only the latest remains.

7.5 Post-exam management

Reads and transforms `.zst` evaluation archives — see *Archive format* for the schema.

7.5.1 `sre cat [options] <file…>`

Prints archive contents. Without a field option, all fields are shown.

Option	Shows
`--data`	`data_json` (serialized `Data`)
`--tests`	Raw test results per machine
`--errors`	Evaluation errors
`--answers`	Student answers
`--grades`	Grade list, total grade, total max
`--files`	Files saved in the archive
`--extract-files`	Extract saved files to the current directory
`--json`	Emit a single JSON dict per file on stdout

7.5.2 `sre check-eval [-s <srelab>] <file…>`

Diffs stored grades against a re-grade (same logic as `re-eval` but without writing files). Prints per-element changes plus a summary count. `-s` selects an updated `srelab.py` or directory; if omitted, uses each archive's `.private/srelab` symlink.

7.5.3 `sre re-eval -s <srelab> -p <prefix> [-d <outdir>] [-r] <file_or_dir…>`

Re-grades archives with an updated `srelab.py`. Useful for fixing a grading bug post-exam.

Option	Description
`-s` / `--srelab`	Updated `srelab.py` or directory.
`-p` / `--prefix`	Prepended to each output archive filename.
`-d` / `--output-dir`	Output directory (default: current).
`-r` / `--recursive`	Recurse into subdirectories.

7.5.4 `sre sheet -o <output.ods> [-r] <file_or_dir…>`

Exports archives to a LibreOffice ODS spreadsheet. Per distinct lab name, three sheets are produced:

Sheet	Content
`{lab_name}`	One row per archive: `login`, `fullname`, `email`, `hostname`, `eval_date`, `errors`, `total_grade`, `total_max`, `mark`, `maximum_mark`, then one column per grade element
`Questions {lab_name}`	Per-question: `max_grade`, `maximum grade`, `average`, `number of projects`, `projects grade 0`
`Sessions {lab_name}`	Per-student best score: `login`, `hostname`, `max score`, `sum of maxima`, then best score per question

`fullname` and `email` come from the archive's `answers` (empty if absent).

7.5.5 `sre outline [-o <summary.ods>] [-d <pdf_dir>] [-r] [--lang <lang>] [--no-timeline] [--remaining-time] [--users-file <file>] <file_or_dir…>`

Generates per-student PDF reports plus a summary ODS — best-graded archive per student. At least one of `-o` / `-d` is required.

Option	Description
`-o` / `--output-file`	Output `.ods` summary (omit to skip).
`-d` / `--pdf-directory`	Output dir for PDFs (omit to skip).
`-r` / `--recursive`	Recurse into subdirectories.
`--lang <lang>`	Force PDF language (e.g. `en`, `fr`).
`--no-timeline`	Omit the evaluation-history table.
`--remaining-time`	Include the time-remaining column in the history.
`--users-file <file>`	User list `LOGIN NAME EMAIL` (whitespace/CSV; `#` comments; <3-field lines skipped).

Name / email resolution: `--users-file` (if provided) → `fullname` / `email` from archive answers. Columns appear only when at least one student has a non-empty value.

7.6 Internal commands (GUI-only)

7.6.1 `sre list [--with-titles]`

Prints a JSON array of lab names relative to `/opt/sre/lab/` (`s4/tp_ssh`, `s2/lab.py`, …). Labs whose components match `params.exam_only_affix` (`_EXAM_`, `_OLD_`, `_DRAFT_`, `_TESTS_`) are hidden.

With `--with-titles`, output becomes `[{"name": <path>, "title": <dict|null>}, …]` — titles come from per-directory `titles.json` (see `sre make-titles`); stale entries are dropped, and labs missing a title get `"title": null` (GUI then falls back to the filename).

7.6.2 `sre export <running_lab> [--sep <N>] [--curved] [--shapes] [--reverse] [--random-seed <N>]`

Exports a running project as a Kathara zip archive, base64-encoded on stdout. Flags tune the embedded schema: `--sep` 0–9 (default 3), `--curved` for curved edges, `--shapes` for geometric nodes, `--reverse` to flip insertion order, `--random-seed` for node-order permutation.

7.6.3 `sre pre-start-exam`

Stops non-exam projects, then starts all exam labs so images are pre-pulled. Called by the GUI ~60 s before `start_after`.

7.6.4 `sre start-exam`

Stamps `started_at` in `exam.json`. Called by the GUI once `now  start_after`.

7.6.5 `sre eval-exam`

Evaluates every running exam project concurrently. Called by the GUI on a periodic timer (every `eval_interval`) and once at exam end.

7.6.6 `sre end-exam`

Stamps `ended_at` and runs a final concurrent eval on every running exam project. Called by the GUI when the exam ends.

CHAPTER

EIGHT

EXAM REFERENCE

8.1 `exam.json` — single source of truth

Lives at `/var/lib/sre/exam.json` (`params.sre_pub_dir + "/" + params.exam_json_name`). The same file must exist on every student PC — distribute via NFS, rsync, or any deployment mechanism. All field names are exposed as `params.exam_*` constants; never hard-code them.

Field	Constant	Type	Meaning
`labs`	`params.exam_labs`	`[[lab, flavor|null], ...]`	Authorised labs + optional flavor preset (required)
`start_afte`	`params.exam_start_after`	ISO datetime	Exam opens after this moment
`end_before`	`params.exam_end_before`	ISO datetime	Exam closes before this moment
`duration`	`params.exam_duration`	int (min)	Max duration; default `params.default_exam_duration = 90`
`eval_inter`	`params.exam_eval_interva`	int (s)	Auto-eval period; default `params.default_eval_interval_during_exams = 60`
`record_ses`	`params.exam_record_sessi`	bool	Record terminal sessions; default `true`
`pre_start_`	`params.exam_pre_start_da`	list[ISO]	Written by `pre-start-exam`; one entry per call
`started_at`	`params.exam_started_at`	ISO datetime	Written by `start-exam`
`ended_at`	`params.exam_ended_at`	ISO datetime	Written by `end-exam`

The first six are set by the instructor (via `sre set-exam`); the last three are runtime state markers.

`labs` format: each entry is a two-element list `[lab_cli_arg, flavor_or_null]`, where `lab_cli_arg` is what `sre start` would receive (e.g. `s4/tp_ssh`) and the second element is the name of a `Flavor` preset or `null`:

```
"labs": [["s4/tp_ssh", null], ["s4/tp_dhcp", "hard"]]
```

Legacy plain-string entries are still accepted on read; always unpack via `params.parse_lab_entry(entry)`.

8.2 GUI exam logic

Each student GUI reads `exam.json` every second (`_update_exam_state`) and runs the same logic — there is no coordinator. The phase is computed from `exam.json` content alone (`_compute_exam_phase`):

- **Waiting** — `start_after` set, `now < start_after`, `started_at` not set.
- **Ended** — `end_before` set and `now  end_before`, **or** `started_at` set and `now  started_at + duration` (duration defaults to `params.default_exam_duration = 90`). The duration check fires only when `started_at` is set — without it, the exam was never formally started.
- Otherwise: **Active**.

`real_starting_time` (drives the countdown) = `started_at` if set, else `start_after` if set, else GUI process start time.

Phas	Preconditions	Actions
Wait ing	—	Show SRE logo + countdown to `start_after`. Once `now  start_after - params.max_duration_between_exam_pre_start_and_start` (60 s) **and** `pre_start_date` is absent, fire `sre pre-start-exam` once (stops non-exam projects, starts exam labs, appends to `pre_start_date`).
Ac- tive	Exam containers up — one running project per lab in `labs`, no others (`_projects_ready`)	Fire `sre start-exam` once (stamps `started_at`; calls `pre-start-exam` internally if `pre_start_date` is missing). Show tabs + countdown. Every `eval_interval` s, fire `sre eval-exam` (skipped if previous still running). If `labs` changes, re-fire `sre pre-start-exam`.
Ende	—	Fire `sre end-exam` once (stamps `ended_at`, runs a final concurrent eval with `save-records --force`). Show "That's All Folks" until the instructor runs `sre del-exam`.

8.3 Commands

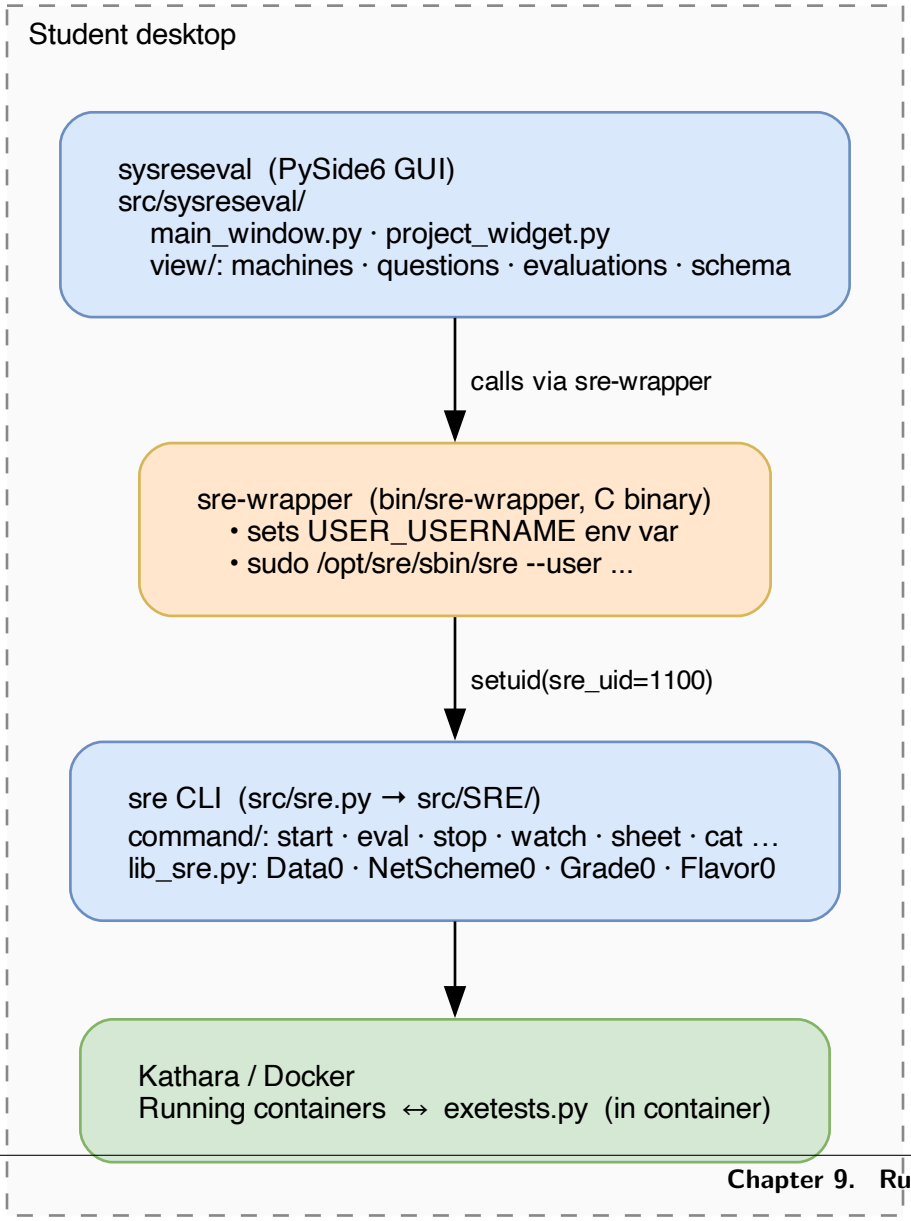
See *CLI Reference* for `set-exam`, `del-exam`, `pre-start-exam`, `start-exam`, `eval-exam`, `end-exam`, `save-records`, `watch`, `cat`, `check-eval`, `re-eval`, `sheet`, `outline`.

CHAPTER

NINE

RUNTIME & INTERNALS

9.1 Architecture



Every command that touches Docker runs as user `sre` (uid `1100`): `sre-wrapper` exports `USER_USERNAME` and calls `sudo sre --user`; the CLI then drops from root to `sre_uid`. How the drop happens depends on the project:

- **Non-privileged projects** (no machine has `privileged=True`): privileges are dropped **permanently** at start via `setuid(sre_uid)` / `setgid(docker_gid)` — the process can never regain root for the rest of its life.
- **Privileged projects** (at least one machine has `privileged=True`): privileges are dropped **temporarily** via `seteuid`/`setegid` only. Kathara requires the deploy/connect/exec paths to run as genuine root (to mount cgroups, attach a TTY to `/sbin/init`, etc.), so the CLI raises the effective uid back to 0 around those operations and lowers it again afterwards. Host-side subprocesses still drop to `sre_uid` in the child via `preexec_drop_to_sre()`, so user code on the host never runs as root.

9.2 Filesystem layout

Path	Description
`/opt/sre/`	Installation root
`/opt/sre/sbin/sre`	CLI binary
`/opt/sre/bin/sre-wrapper`	C binary for student use (handles sudo)
`/opt/sre/bin/sysreseval`	GUI launcher script
`/opt/sre/lab/`	Lab files
`/opt/sre/lib/`	Shared libraries for project files (`ips.py`, `std.py`, `exetests.py`, `net_config.py`, `dhcp.py`, `ping.py`, `ssh.py`, `tls.py`, `grade_helpers.py`, `frr.py`, `state_helpers.py`, `pcap_gen.py`, `utils.py`)
`/opt/sre/graphics/`	SVG icons and logos
`/opt/sre/locale/`	Compiled gettext translations for the CLI
`/var/lib/sre/`	Runtime state root (`sre_pub_dir`)
`/var/lib/sre/exam.json`	Exam configuration (absent = no exam)
`/var/lib/sre/projects/`	One directory per running lab instance
`/var/lib/sre/archives/`	Default evaluation archive directory
`/var/lib/sre/last_self_grade`	Self-grade cooldown timestamps
`/home/sre/`	Shared directories for projects with `shared_path=True` (mode `0o777`, removed on stop/wipe)

9.2.1 Running lab directory

```
/var/lib/sre/projects/{timestamp}@@@{lab_name}@@@{username}/
+-- .private/              mode 0o700
|   +-- data.json          serialized Data instance, mode 0o600
|   +-- srelab             symlink to the lab's srelab.py
```

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```
|   +-- eval_in_progress    lock file (PID) during active eval
|   +-- auto_eval.log       one ISO timestamp per student self-evaluation
+-- info.json               public machine/question metadata (InfoLab)
+-- scheme.svg              graphviz network diagram
+-- answers/
|   +-- answers.json        student answers (updated by GUI)
|   +-- cheat.json          instructor-provided answers (if any)
+-- shared/                 mode 0o777 (only if shared_path=True)
```

9.3 Configuration

9.3.1 Constants (`src/SRE/params.py`)

Constant	Value	Description
`sre_uid`	`1100`	UID of the `sre` system user
`docker_gid`	`988`	GID of the `docker` group
`max_docker_concurrency`	`16`	Max parallel Docker API calls during eval
`default_eval_interval_during_exams`	`60`	Default auto-eval period (s)
`default_exam_duration`	`90`	Default exam duration (min), used if `duration` absent from `exam.json`
`default_inactivity_threshold_in_watc`	`90`	`sre watch` inactivity alert threshold
`max_duration_between_exam_pre_start_`	`60`	Window before `start_after` during which `pre-start-exam` fires (s)
`email_in_gecos_last_field`	`True`	Extract last GECOS sub-field as student email
`terminal_cmd_prefix`	`["/usr/bin/ mate-terminal", "--"]`	External terminal emulator
`terminal_font_size`	`12`	Default terminal font size (pt)
`terminal_color_scheme`	`"black_on_white"`	Default terminal colors
`exam_only_affix`	`["_EXAM_", "_OLD_", "_DRAFT_", "_TESTS_"]`	Name substrings that hide labs from `sre list`
`authorized_src_dir`	`['/opt/sre/lab', '/ home/etudiant']`	Allowed lab source directories
`hostname_keyword` / `login_keyword` / `fullname_keyword` / `email_keyword` / `language_keyword`	`"hostname"` / `"login"` / `"fullname"` / `"email"` / `"language"`	Keys used in `answers.json` and archive `answers` dict

9.3.2 Environment variables

GUI / wrapper → sre CLI

Variable	Used by	Description
`USER_USERNAME`	sre CLI (with `--user`)	Student login name (set by `sre-wrapper` from the invoking `USER`)
`SRE_PUB_DIR`	sre CLI, GUI, tests	Override `/var/lib/sre`
`SRE_WRAPPER`	params.py	Override the sre-wrapper path
`LANG`	sre CLI	Loads French help strings when set to a French locale

sre CLI → containers

Injected into machines at deploy / `exec` / state-transition time.

Variable	Description
`SRE_LAB_NAI`	Running lab name (`{timestamp}@@@{lab_name}@@@{username}`) — defined by `params.sre_name_env_variable`
`SRE_XAUTH_(`	X authority cookie forwarded into privileged containers so GUI apps can reach the host X server — defined by `params.sre_xauth_cookie_env_variable`
`SRE_HOST_II`	Host IP reachable from containers (default `172.17.0.1`) — defined by `params.sre_host_ip_env_variable`

9.3.3 Locale

The CLI uses `gettext` (`locale/fr/LC_MESSAGES/sre.mo`); the GUI uses Qt `.qm` catalogs (`translations/sysreseval_fr.qm`). Both are regenerated by `make translations`.

9.4 Dynamic module loading

Each lab is loaded at runtime via `importlib.util.spec_from_file_location('srelab', ...)`. The module path must sit under `params.authorized_src_dir` (`/opt/sre/lab/` or `/home/etudiant/`), and `/opt/sre/lib/` is prepended to `sys.path` so labs can `from ips import …`, `from net_config import …`, etc.

9.5 Data serialization

`Data0` encodes to JSON/msgpack with a polymorphic envelope:

```
{
  "__type__": "srelab.Data",
  "data": {
    "secret": "changeme",
    "ips":  {"router": "10.0.0.1/24"},
    "nets": {"lan1": "10.0.0.0/24"}
  }
}
```

Value	Encoding
`IPv4Interface` inside `ips` container	`"10.0.0.1/24"`
`IPv4Network` inside `nets` container	`"10.0.0.0/24"`
`IPv4Address` (standalone field)	`{"__ip__": "10.0.0.1"}`
`IPv4Network` (standalone field)	`{"__net__": "10.0.0.0/24"}`
Nested `Data0` subclass	`{"__type__": "module.ClassName", "data": {...}}`

`data.json` is saved mode `0o600` inside `.private/` (mode `0o700`).

9.6 Test execution

`Grade.run_tests()` per step:

1. Call `grade()` to *register* tests (no execution).
2. Group commands per machine and encode them into one env var `EXETESTS@@@{timeout}:{cmd}@@@{timeout}:{cmd}@@@…`.
3. Run `exetests.py` inside each container; outputs are separated by a per-run UUID.
4. Parse outputs into `self._tests[(machine, step)][(command, timeout)]`; repeat until `step > max_step`; call `grade()` a final time to compute grades.

Up to 16 machines tested concurrently (`ThreadPoolExecutor`). Exit codes: `0`–`255` actual, `-1` = timeout.

9.7 Progress reporting

`sre start` emits JSON lines to **stderr** so the GUI can render progress:

```
{"phase": "pull",   "status": "start",    "image": "sre/base:1.10"}
{"phase": "pull",   "status": "progress", "image": "sre/base:1.10", "percent": 42}
{"phase": "pull",   "status": "end",      "image": "sre/base:1.10"}
{"phase": "deploy", "status": "start",    "total": 3}
{"phase": "deploy", "status": "progress", "current": 1, "total": 3}
{"phase": "deploy", "status": "end",      "total": 3}
```

9.8 Security model

Concern	Mechanism
Allowed UIDs	`sre.py` checks `os.geteuid()` against `sre_uid` and 0; exits otherwise
Privilege drop	Root calls `_drop_permanently()`, which `sys.exit()`s on failure
Lab path whitelist	`utils.py` checks `current_srelab_file` is under `authorized_src_dir`
Eval lock	Lock file opened with `O_CREAT | O_WRONLY | O_NOFOLLOW`; `flock(LOCK_EX)` prevents concurrent evals and symlink attacks
Host commands drop privileges	`preexec_drop_to_sre()` is `preexec_fn` for every host-side `subprocess.run()` (eval, state transitions)
Shell injection	`tls.py` quotes with `shlex.quote()`; `state_helpers.py` writes passwords to `/tmp/.sre_chpasswd` (mode `0o600`); usernames passed via env vars
`data.json` secrecy	Mode `0o600` in `.private/` (mode `0o700`)
Unknown `__type__` deserialization	`Data0.from_dict()` / `unpack()` / `Flavor0` raise `ValueError`
Single GUI instance	PID file at `/tmp/sysreseval-{uid}.pid` kills the previous instance
Student command restrictions	`user_not_allowed()` / `user_not_allowed_in_exam_mode()`
State access control	`@sre_state(user_allowed=False)` blocks students from re-applying protected states

9.9 Archive format

Every evaluation writes a `zstandard`-compressed `msgpack` file with a `.zst` extension to `params.archive_dirs` (default `['/var/lib/sre/archives']`; labs may extend the list via a module-level `archive_dirs` attribute).

Filename: `{eval_date}_{running_lab_name}.zst` — `eval_date` is `YYYYmmddHHMMSS`, `running_lab_name` is `{timestamp}@@@{lab_name}@@@{username}`. The leading timestamp sorts chronologically with `ls`.

```
archive = {
    "running_lab_name": "{timestamp}@@@{lab_name}@@@{username}",
    "eval_date":        "20260615110042",
    "data_json":        "<JSON string>",        # serialized Data
    "tests": {
        ("router", 1): {("ping -c1 8.8.8.8", 5): ("64 bytes ...\n", 0), ...},
        ...
    },
    "errors":     ["error message", ...],
    "answers": {
        "<question_hash>":     "<student answer>",
        "hostname":            "pc-101",
        "login":               "alice",
        "fullname":            "Alice Martin",      # from GECOS, always present
        "email":               "alice@example.com", # when email_in_gecos_last_field=True
        "language":            "fr",
        "answers_updated_at":  "20260615110000",
        "auto_eval_count":     2,                   # self-evals before this one
        "exam_mode":           true,                # exam-mode only
        "exam_started_at":     "2026-06-15T09:00",  # exam-mode only
        "exam_duration":       120,                 # exam-mode only
```

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```
        "exam_time_remaining": 3600,                # exam-mode only
    },
    "grade_list": [
        {"title": "Connectivity", "max_grade": 4, "grade": 4,
         "grade_letter": null, "description": ""},
        ...
    ],
    "total_grade":  14.0,
    "total_max":    20.0,
}
```

Archives are written by `sre eval`, `eval-all`, `eval-exam`, `end-exam`, `re-eval`, and consumed by `sre cat`, `sheet`, `outline`, `check-eval`, `watch` — see *CLI Reference*.